



Introduction to Information Systems

Lecture 1

Information Systems: An Overview



Dr. Mahmoud Shuker Mahmoud
mahmoud.shukur@mtu.edu.iq

Why Learn About Information Systems?

- Information system (IS):
 - A set of interrelated components that collect, manipulate, and disseminate data and information and provide feedback to meet an objective
- Businesses:
 - Can use information systems to increase revenues and reduce costs
- Information systems used by:
 - Sales representatives
 - Managers
 - Financial advisors
- Information systems:
 - Indispensable tools to help you achieve your career goals

Information Concepts

Data, Information, and Knowledge

- **Data:**
 - Raw facts
- **Information:**
 - Collection of facts organized in such a way that they have value beyond the facts themselves
 - One of an organization's most valuable resources
 - Often confused with the term *data*
- **Knowledge:**
 - Awareness and understanding of a set of information

Data, Information, and Knowledge (continued)

Data	Represented by
Alphanumeric data	Numbers, letters, and other characters
Image data	Graphic images and pictures
Audio data	Sound, noise, or tones
Video data	Moving images or pictures

Table 1.1

Types of Data

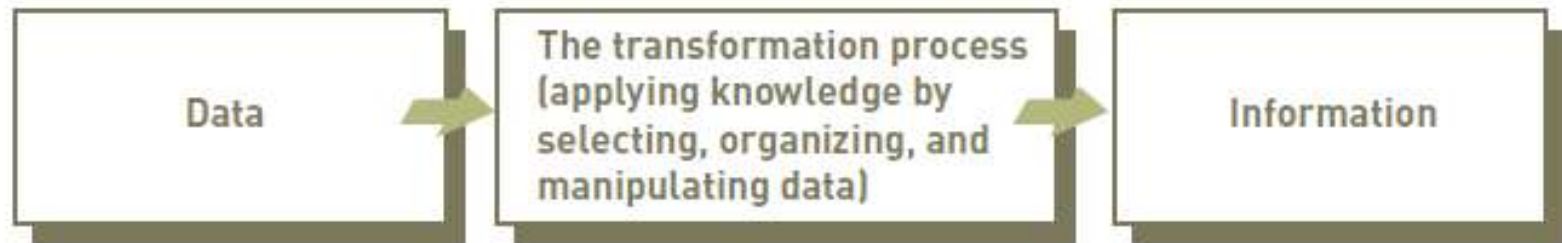


Figure 1.2

The Process of Transforming
Data into Information

The Characteristics of Valuable Information

- If an organization's information is not accurate or complete:
 - People can make poor decisions, costing thousands, or even millions, of dollars
- Depending on the type of data you need:
 - Some characteristics become more important than others

The Value of Information

- Directly linked to how it helps decision-makers achieve their organization's goals
- Valuable information:
 - Can help people and their organizations perform tasks more efficiently and effectively

The Characteristics of Valuable Information (continued)

Characteristics	Definitions
Accessible	Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs.
Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process. (This is commonly called garbage in, garbage out [GIGO].)
Complete	Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it.
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the total value the company has invested in inventory.
Relevant	Relevant information is important to the decision maker. Information showing that lumber prices might drop might not be relevant to a computer chip manufacturer.
Reliable	Reliable information can be trusted by users. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up might not be reliable.
Secure	Information should be secure from access by unauthorized users.
Simple	Information should be simple, not overly complex. Sophisticated and detailed information might not be needed. In fact, too much information can cause information overload, whereby a decision maker has too much information and is unable to determine what is really important.
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.

Table 1.2

System Concepts

- System:
 - Set of elements or components that interact to accomplish goals
- Components of a system:
 - Inputs
 - Processing mechanisms
 - Outputs
 - Feedback

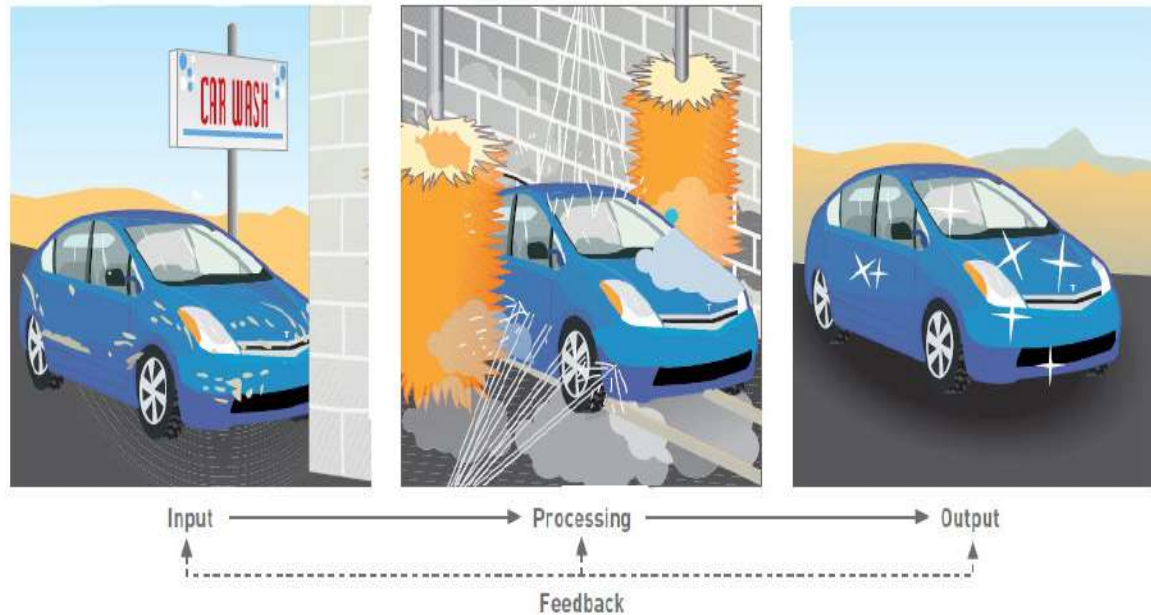


Figure 1.3

Components of a System

A system's four components consist of input, processing, output, and feedback.

System Performance and Standards

- **Efficiency:**
 - Measure of what is produced divided by what is consumed
- **Effectiveness:**
 - Measure of the extent to which a system attains its goals
- **System performance standard:**
 - Specific objective of the system

System Performance and Standards (continued)

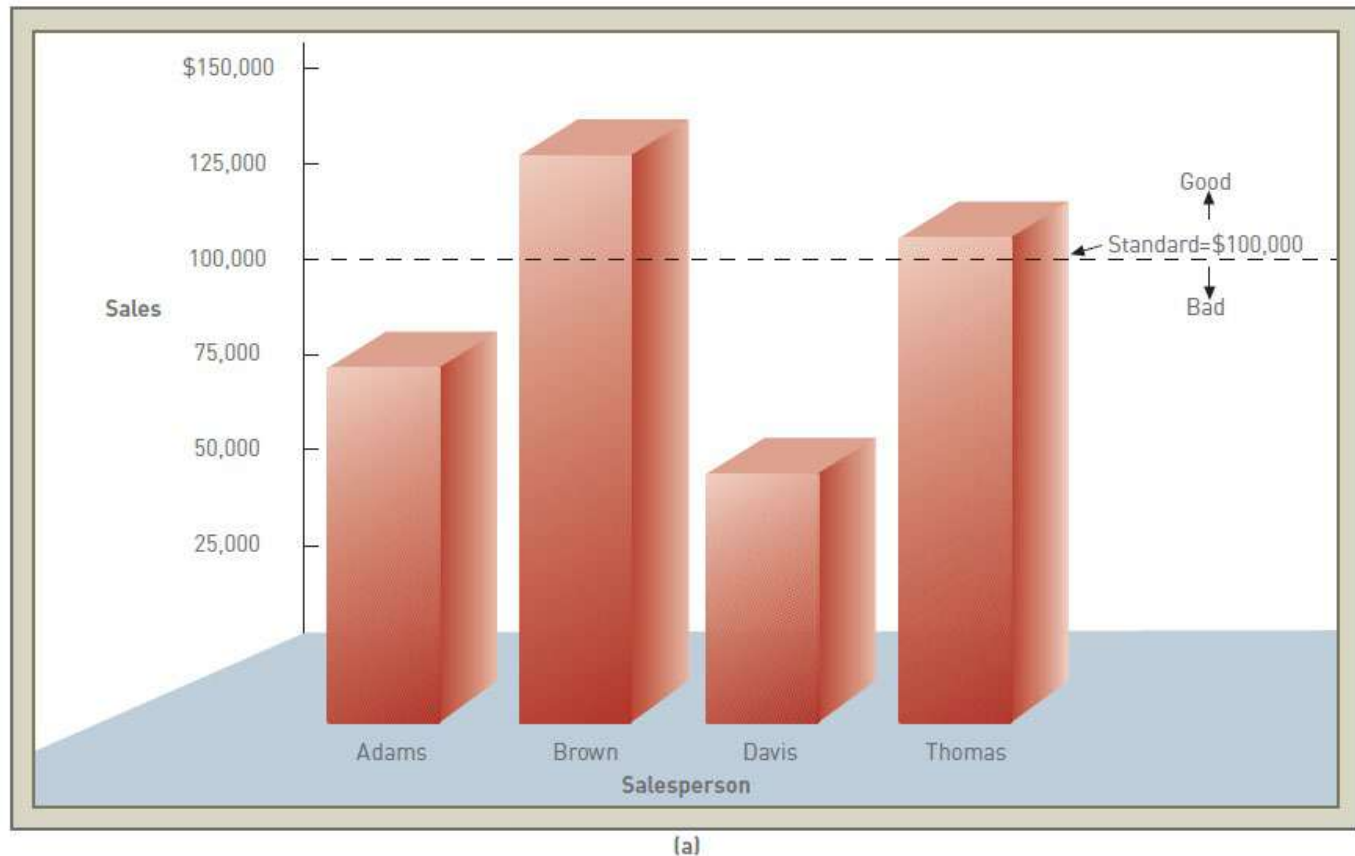
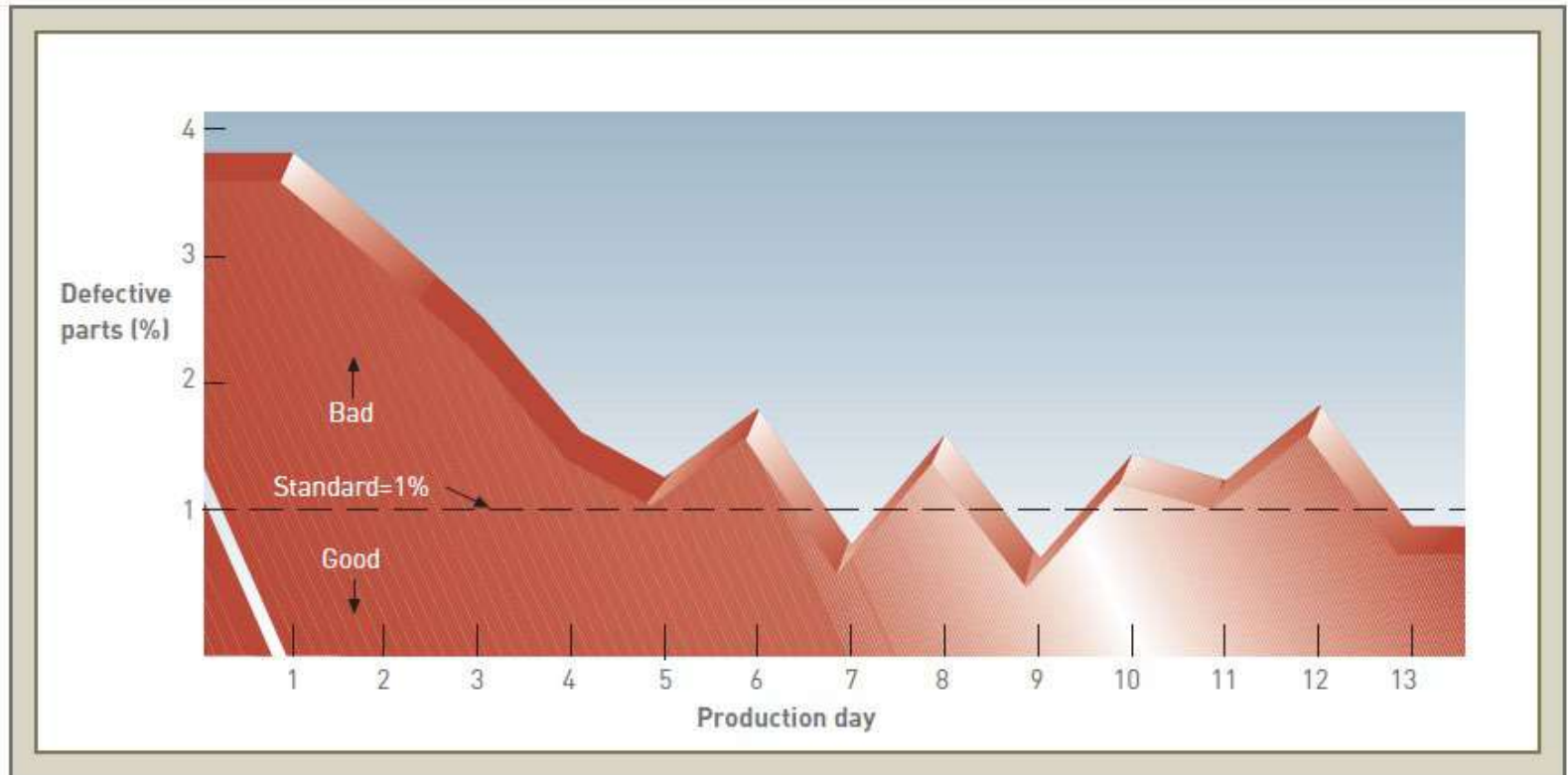


Figure 1.4

System Performance
Standards

System Performance and Standards (continued)



(b)

Figure 1.4

System Performance
Standards

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What is an Information System?

- Information system (IS) is a set of interrelated elements that:
 - Collect (input)
 - Manipulate (process)
 - Store
 - Disseminate (output) data and information
 - Provide a corrective reaction (feedback mechanism) to meet an objective



Figure 1.5

The Components of an
Information System

Feedback is critical to the
successful operation of a system.

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Input, Processing, Output, Feedback

- **Input:**
 - Activity of gathering and capturing raw data
- **Processing:**
 - Converting data into useful outputs
- **Output:**
 - Production of useful information, usually in the form of documents and reports
- **Feedback:**
 - Information from the system that is used to make changes to input or processing activities

Computer-Based Information Systems

- **Technology infrastructure:**
 - Includes all hardware, software, databases, telecommunications, people, and procedures
 - Configured to collect, manipulate, store, and process data into information

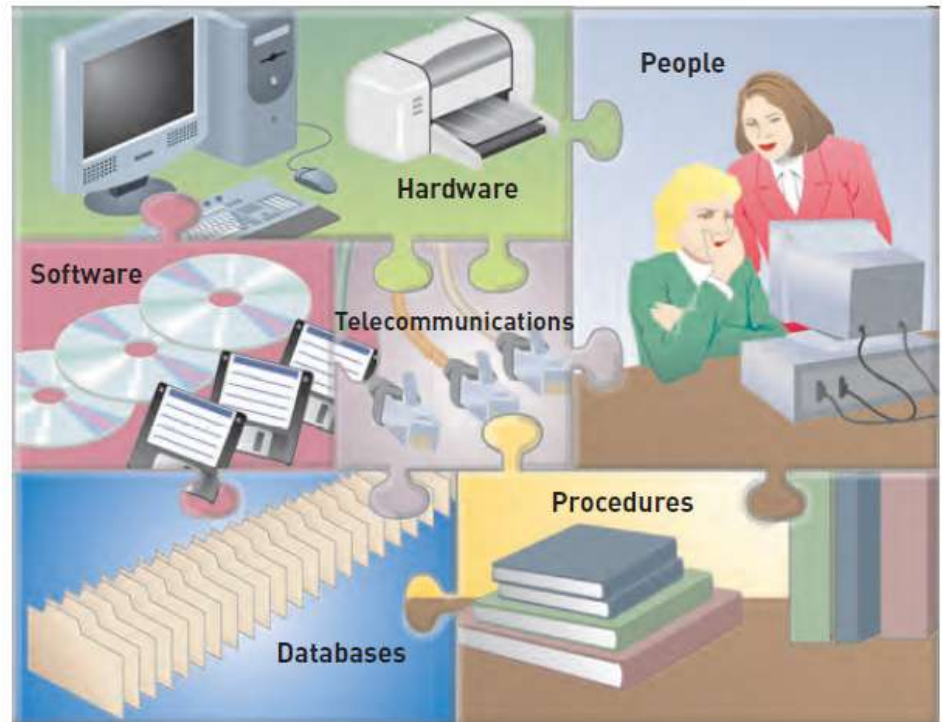


Figure 1.6

The Components of a
Computer-Based Information
System.

Computer-Based Information Systems (continued)

- **Hardware:**
 - Consists of computer equipment used to perform input, processing, and output activities
- **Software:**
 - Consists of the computer programs that govern the operation of the computer
- **Database:**
 - Organized collection of facts and information, typically consisting of two or more related data files
- **Telecommunications, networks, and the Internet**
 - The electronic transmission of signals for communications

Computer-Based Information Systems (continued)

- **Networks**
 - Connect computers and equipment to enable electronic communication
- **Internet**
 - World's largest computer network, consisting of thousands of interconnected networks, all freely exchanging information
- **Intranet:**
 - Internal network that allows people within an organization to exchange information and work on projects
- **Extranet:**
 - Network that allows selected outsiders, such as business partners and customers, to access authorized resources of a company's intranet
- **People:**
 - The most important element in most computer-based information systems
- **Procedures:**
 - Include strategies, policies, methods, and rules for using the CBIS



Introduction to Information Systems

Lecture 2

Hardware: Input, Processing, and Output Devices



Dr. Mahmoud Shuker Mahmoud
mahmoud.shukur@mtu.edu.iq

Why Learn About Hardware?

- Organizations invest in computer hardware to:
 - Improve worker productivity
 - increase revenue, reduce costs
 - Provide better customer service
 - Speed up time-to-market
 - Enable collaboration among employees

Principles and Learning Objectives

- Computer hardware must be carefully selected to meet the evolving needs of the organization and its supporting information systems
 - Describe the role of the central processing unit and main memory
 - State the advantages of multiprocessing and parallel computing systems and provide examples of the problems they address.
 - Describe the access methods, capacity, and portability of various secondary storage devices
 - Identify and discuss the speed, functionality, and importance of various input and output devices
 - Identify the characteristics of and discuss the usage of various classes of single-user and multiuser computer systems.
- The computer hardware industry is rapidly changing and highly competitive, creating an environment ripe for technological breakthroughs
- The computer hardware industry and users are implementing green computing designs and products

Computer Systems: Hardware Components

- Central processing unit (CPU)
- Input devices
- Output devices
- Primary storage devices
- Secondary storage devices
- Communications devices

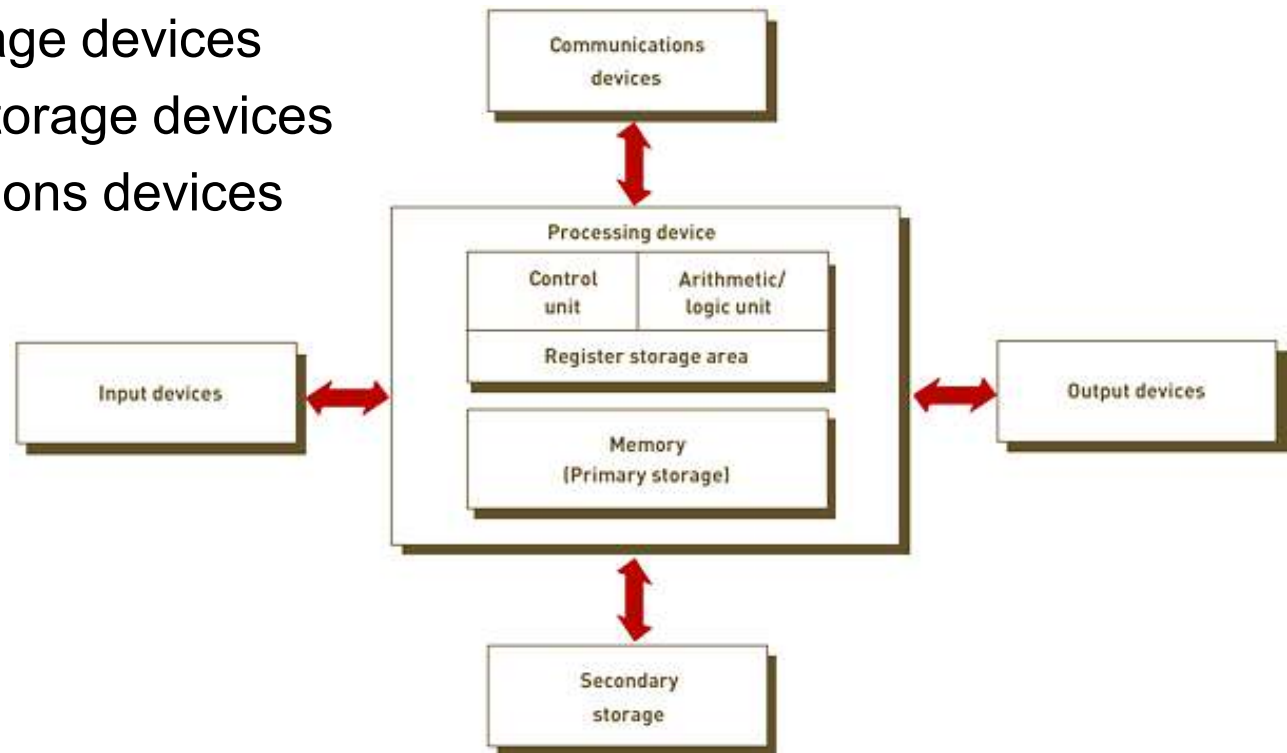


Figure 2.1: Hardware Components

Central processing unit (CPU)

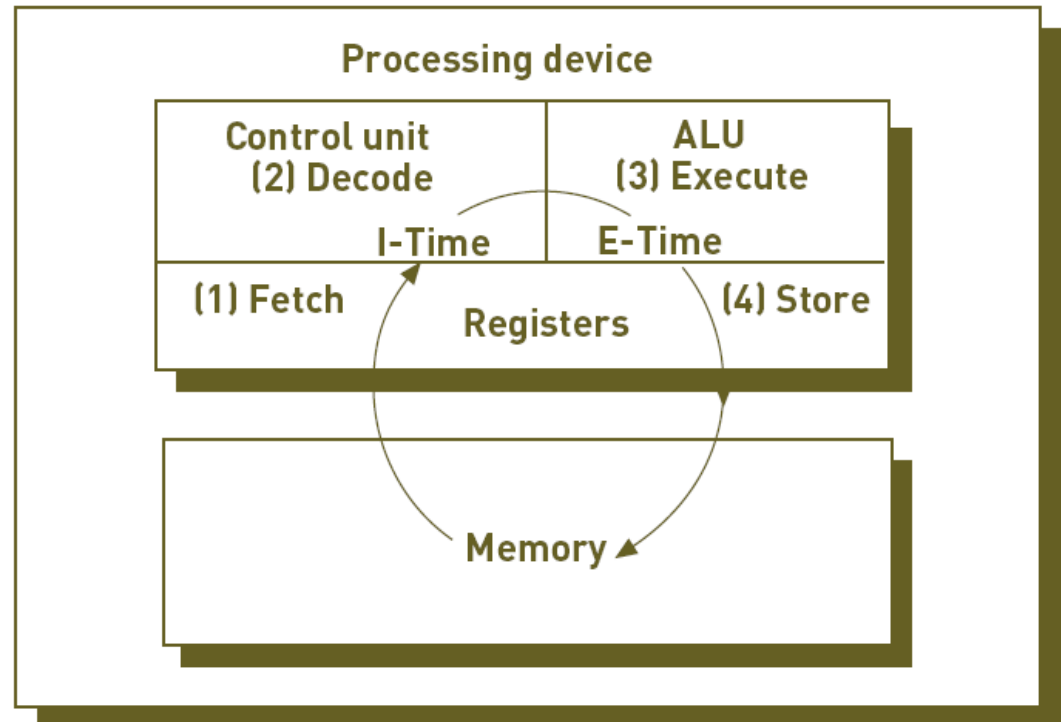
The Central processing unit (CPU) consists of the arithmetic/logic unit, the control unit, and the register areas

- **Arithmetic/logic unit (ALU):** Performs mathematical calculations and makes logical comparisons.
- **Control unit (CU):** Sequentially accesses program instructions, decodes them, and coordinates data flow in and out of the ALU, registers, primary storage, secondary storage, and various output devices.
- **Registers:** High-speed storage areas are used to hold small units of program instructions and data temporarily.
- **Primary storage:** Also called main memory closely associated with the CPU.

Hardware Components in Action

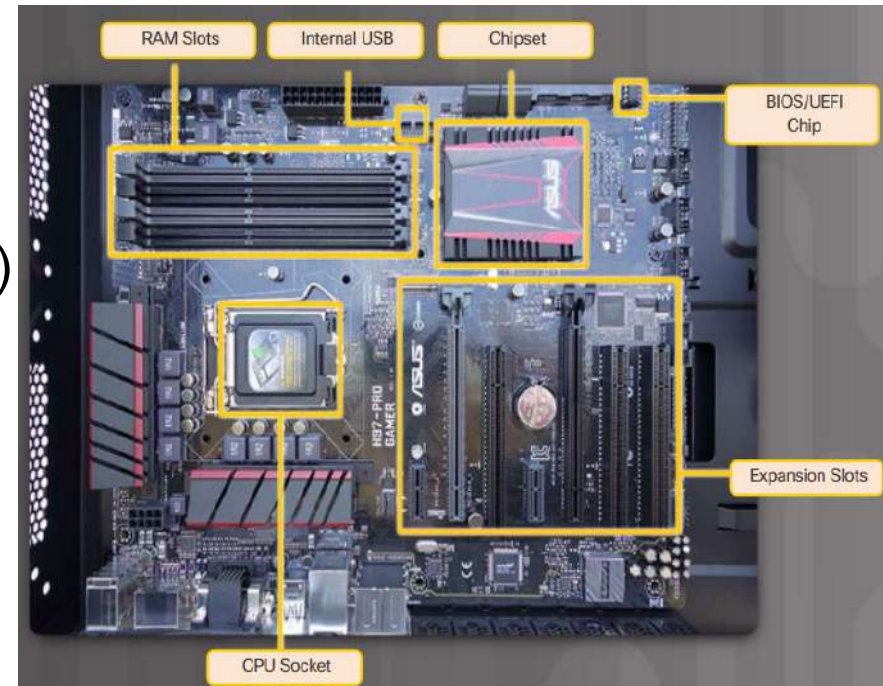
- **Step 1: Fetch instruction**
- **Step 2: Decode instruction**
- **Step 3: Execute instruction**
- **Step 4: Store results**

Figure 2.2: Instruction Cycle operation



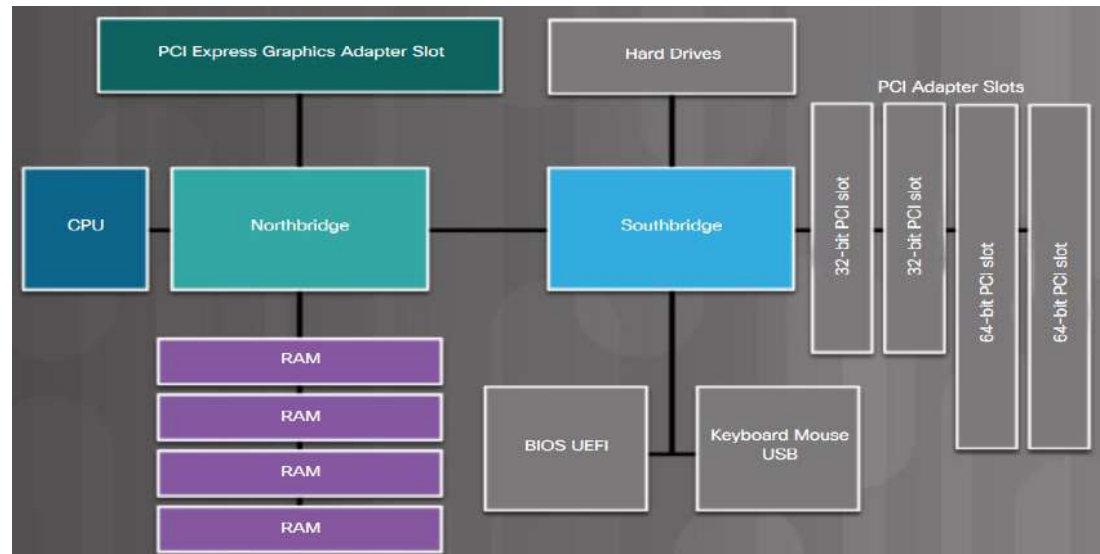
Processing and Memory Devices: Power, Speed, and Capacity

- **System unit (Motherboard)**
 - Houses the components responsible for processing (the CPU and memory)
- **Major components on a motherboard include:**
 - Central Processing Unit (CPU)
 - Random Access Memory (RAM)
 - Expansion slots
 - Chipset
 - Basic input/output system (BIOS)
 - SATA connectors
 - Internal USB connector



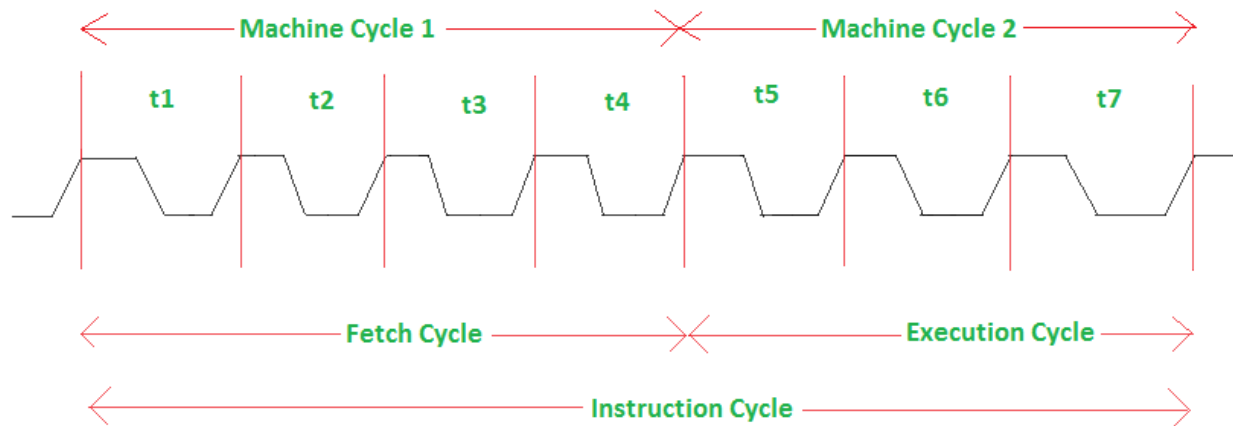
Motherboard Chipset

- **Chipset** consists of the integrated circuits on the motherboard that control how system hardware interacts with the CPU and motherboard.
- Most chipsets consist of the following two types:
 - **Northbridge** – Controls high speed access to the RAM and video card.
 - **Southbridge** – Allows the CPU to communicate with slower speed devices including hard drives, Universal Serial Bus (USB) ports, and expansion slots.



Processing Characteristics and Functions

- **Machine cycle, or instruction cycle**, is a sequence of steps that a CPU goes through in order to execute a single machine language instruction
- Machine cycle time is measured in:
 - Nanoseconds (1 billionth of a second) = (10^{-9})
 - Picoseconds (1 trillionth of a second) = (10^{-12})
 - MIPS (millions of instructions per second)

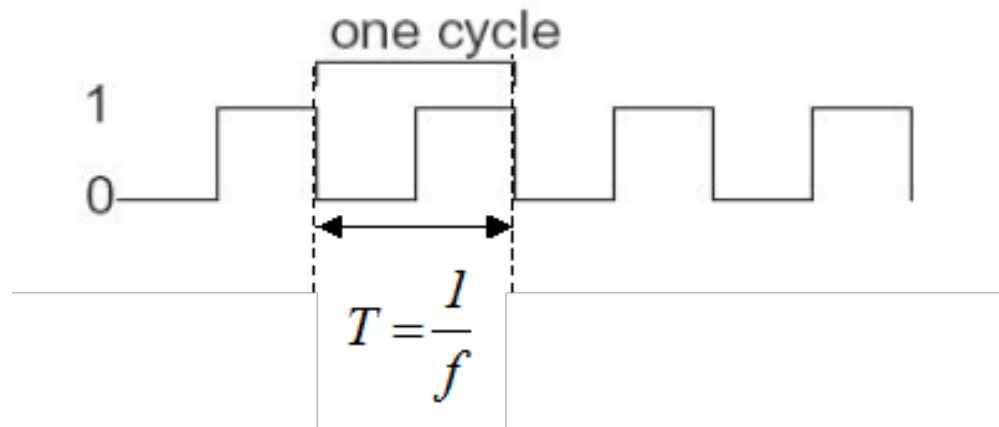


Instruction cycle in 8085 microprocessor

Processing Characteristics and Functions (continued)

- **Clock Speed**

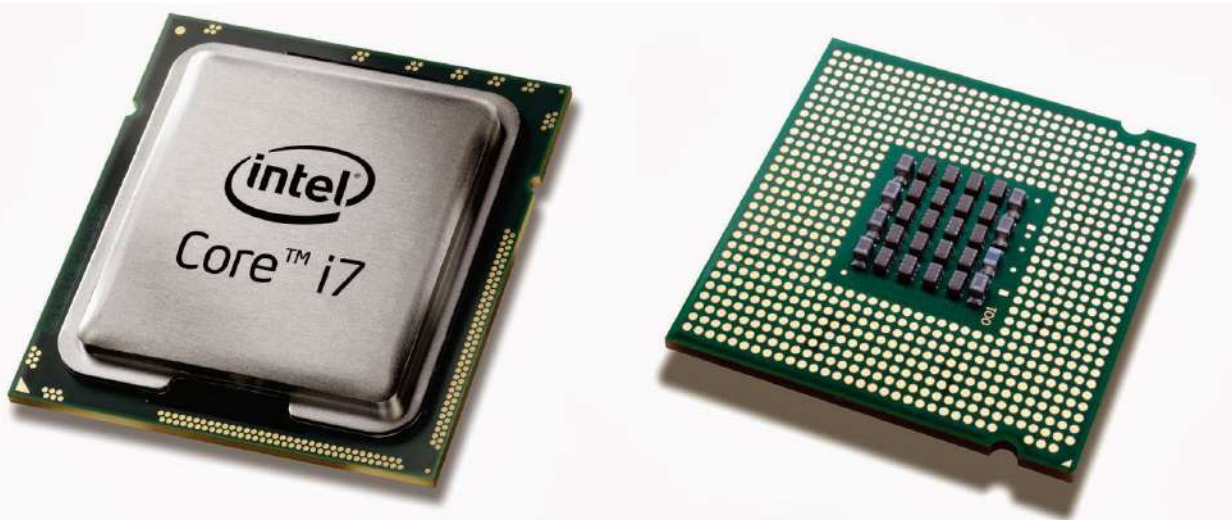
- Series of electronic pulses produced at a predetermined rate that affects machine cycle time
- Often measured in:
 - Megahertz (MHz): millions of cycles per second = (10^6)
 - Gigahertz (GHz): billions of cycles per second = (10^9)



Processing Characteristics and Functions (continued)

Physical characteristics of the CPU:

- Most CPUs are collections of digital circuits imprinted on silicon wafers or chips, each no bigger than the tip of a pencil eraser.



Physical Characteristics of the CPU (continued)

- Moore's Law:

- Hypothesis stating that transistor densities on a single chip will double every two years

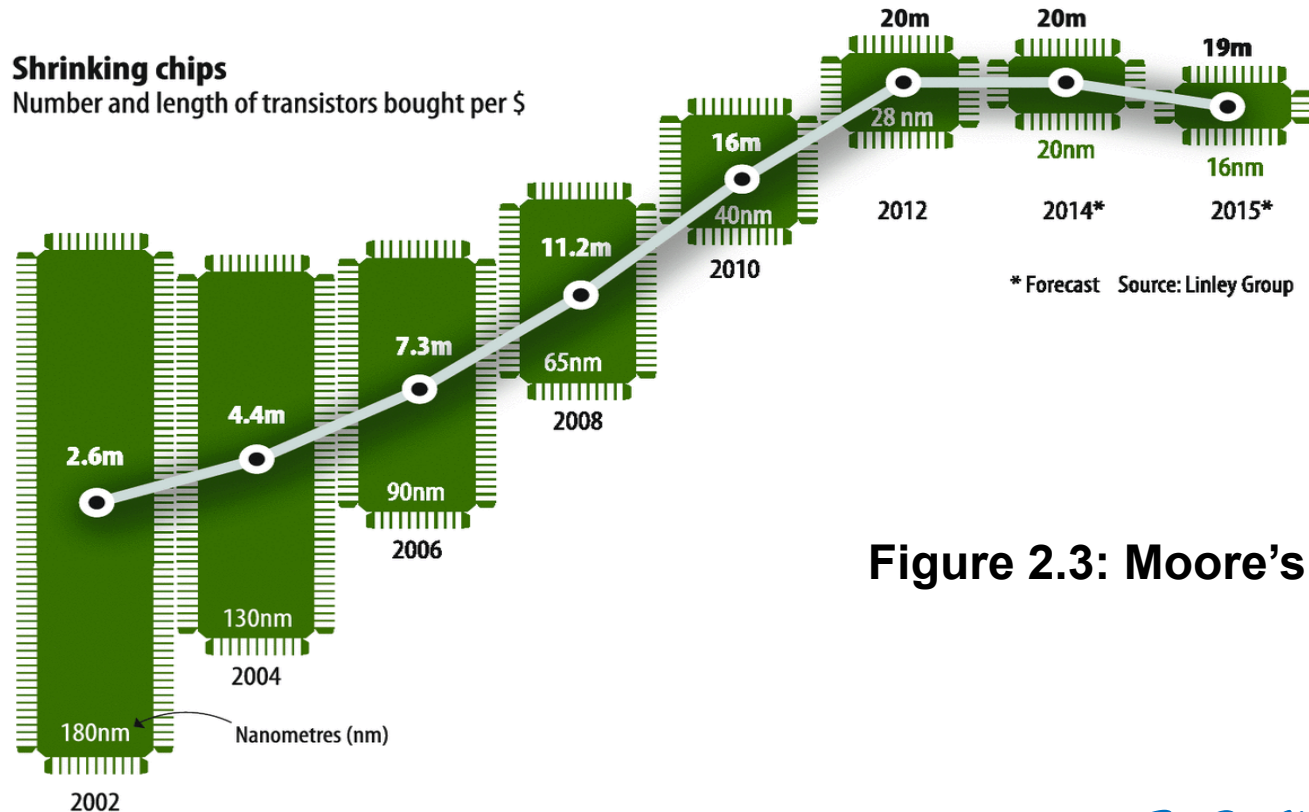


Figure 2.3: Moore's Law

Parallel Computing

- **Parallel computing**
 - Simultaneous execution of the same task on multiple processors to obtain results faster.
- **Cluster & Grid computing**
 - Use a collection of computers to work in a coordinated manner to solve a common problem.
- **Cloud computing**
 - Uses a giant cluster of computers that serves as a host to run applications that require high-performance computing
 - Supports a wider variety of applications than grid computing
 - Pools computing resources so they can be managed primarily by software rather than people

Memory Characteristics and Functions

- **Main memory**
 - Located physically close to the CPU, but not on the CPU chip itself
 - Rapidly provides data and instructions to the CPU
- **Storage Capacity:** Eight bits together form a byte

Name	Abbreviation	Number of Bytes
Byte	B	1
Kilobyte	KB	2^{10} or approximately 1,024 bytes
Megabyte	MB	2^{20} or 1,024 kilobytes (about 1 million)
Gigabyte	GB	2^{30} or 1,024 megabytes (about 1 billion)
Terabyte	TB	2^{40} or 1,024 gigabytes (about 1 trillion)
Petabyte	PB	2^{50} or 1,024 terabytes (about 1 quadrillion)
Exabyte	EB	2^{60} or 1,024 petabytes (about 1 quintillion)

Table (2.1): Computer Storage Units

Types of Memory

- **Read-only memory (ROM):** Provides permanent storage for data and instructions that do not change nonvolatile, such as (ROM, PROM, EPROM, EEPROM)
- **Random access memory (RAM):** the temporary working storage for data and programs accessed by the CPU. RAM is a volatile memory.
- Types of RAM
 - Dynamic RAM (DRAM)
 - Static RAM (SRAM)
 - Synchronous Dynamic RAM (SDRAM)
 - Double Data Rate Synchronous Dynamic RAM (DDR SDRAM)
 - DDR2 SDRAM, DDR3 SDRAM, DDR4 SDRAM.
 - GDDR Synchronous Dynamic RAM (GDDR SDRAM)
 - DDR5



Types of Memory (Cont.)

- The speed of memory has a direct impact on how much data a processor can process in a given period of time.
- The fastest memory is typically static RAM (SRAM), which is used as **cache memory** for storing the most recently used data and instructions by the CPU.

- **Cache memory:** High-speed memory that a processor can access more rapidly than main memory
- The three most common types of cache memory are:
 - **L1 cache** – integrated into the CPU
 - **L2 cache** – was originally mounted on the motherboard but is now integrated into the CPU
 - **L3 cache** – used some high-end workstations and server CPUs

Types of Memory (continued)

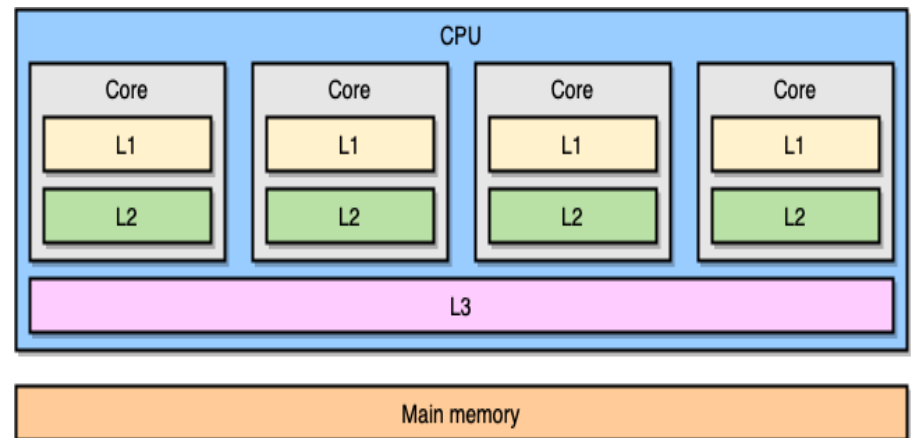
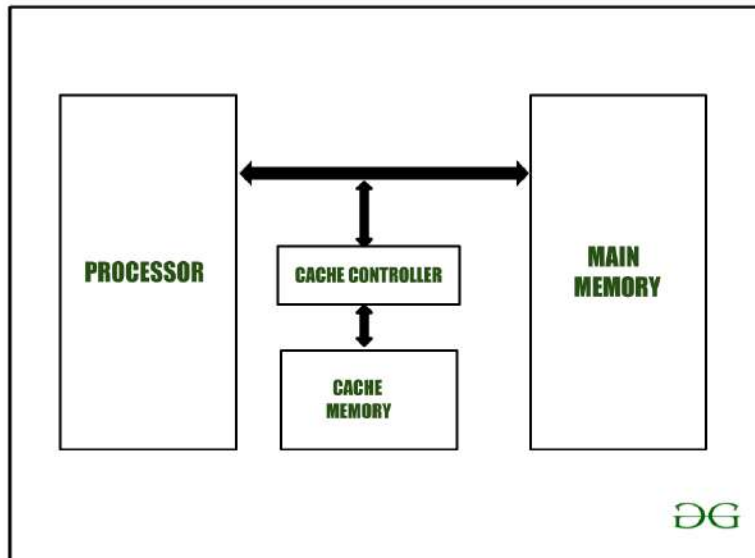


Figure (2.4): Cache Memory

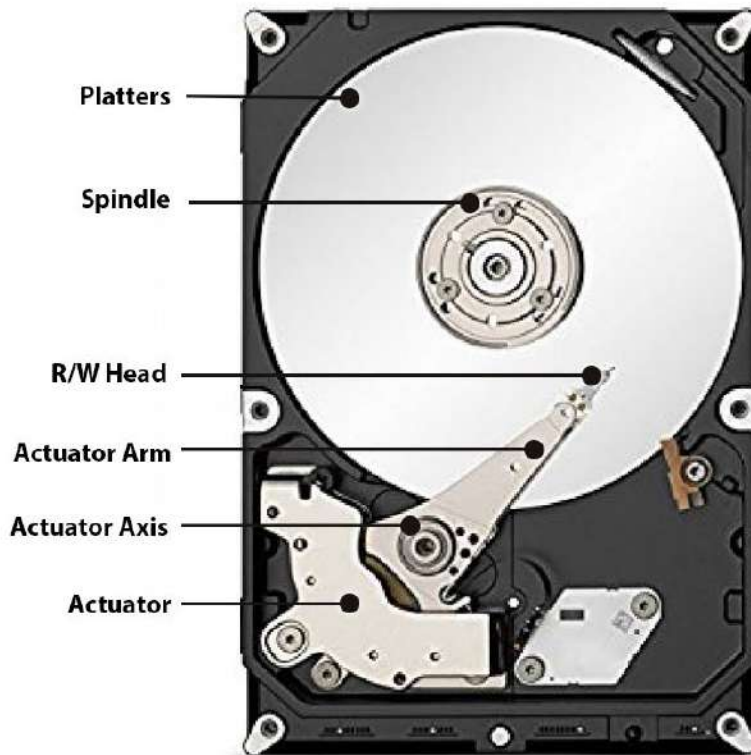
Processors can access this type of high-speed memory faster than main memory. Located on or near the CPU chip, cache memory works with main memory. A cache controller determines how often the data is used, transfers frequently used data to cache memory, and then deletes it when it goes out of use. Cache memory works with main memory, located on or near the CPU chip

Secondary Storage

- Compared with memory, offers the advantages of nonvolatility, greater capacity, and greater economy
- On a cost-per-megabyte basis
 - Most forms of secondary storage are considerably less expensive than primary memory
- Storage media that allow faster access
 - Generally more expensive than slower media
- Access method can be sequential or direct access
 - Sequential access records must be retrieved in the order in which it is stored
 - Direct access records can be retrieved in any order
- The most common types of secondary storage used these days are HDD, SSD, CD, DVD, Flash Memory, and Memory cards.

Secondary Storage (Cont.)

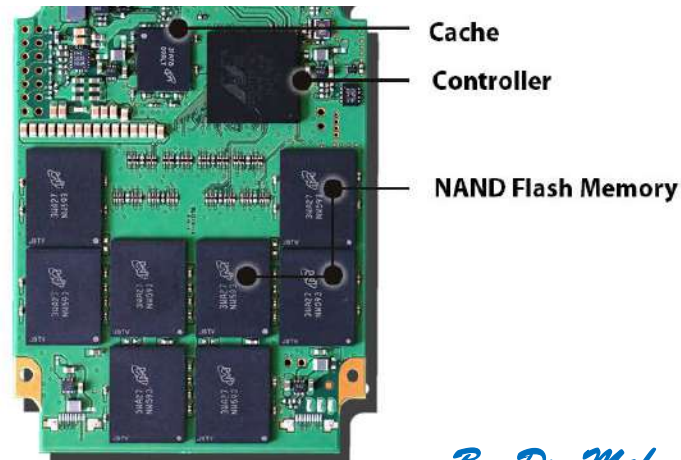
HDD 3.5"



WHAT IS THE DIFFERENCE BETWEEN HDD VS SSD VS NVME?

HDD, SSD, and NVMe are all types of storage devices used in computers. HDD stands for Hard Disk Drive and is a traditional storage device that uses spinning disks to read and write data. SSD stands for Solid State Drive and uses flash memory to store data. NVMe stands for Non-Volatile Memory Express and is a newer protocol used for accessing high-speed storage devices like SSDs. NVMe can provide much faster read and write speeds compared to traditional SATA-based SSDs.

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HDD vs. SSD

Topic	SSD	HDD
Speed	35 to 100 microsecond access speed, nearly 100 times faster than HDD.	Takes about 5000 to 10, 000 microseconds to access data.
Reliability	SSD has no moving parts, so it can keep your data safe when your laptop bag drops or your system gets shaken while it's operating.	The HDD has moving parts and magnetic platters, so they are prone to get wear and tear with more usage.
Noise	Since SSD comes with no moving parts, it won't make much noise when working.	With spinning platters and moving heads, the HDD will produce noises when functioning.
Heat	No moving parts and using the flash memory, the SSD generates less heat, helping increase the lifespan and durability.	With inside moving parts, more heat will be added and damage the electronics gradually. Higher heat will cause more potential damage.
Power	SSD uses less power, so you can enjoy lower engery bill over time and an increased battery life of an laptop.	HDD costs more power than an SSD, because all the HDD parts are required to spin the platters.
Size	Available in 2.5", 1.8", and 1.0", helping save lots of precious space in computers, especially desktops.	Usually, HDD has only 3.5"and 2.5" size for desktops and laptops respectively, no smaller ones.

Table (2.2): Comparison between HDD and SSD

Enterprise Storage Options

- Network-attached storage (NAS)
 - It is a dedicated file storage that enables multiple users and heterogeneous client devices to retrieve data from a centralized disk capacity.
- Storage area network (SAN)
 - It is a computer network that provides access to consolidated, block-level data storage. SANs are primarily used to access data storage devices, such as disk arrays and tape libraries, from servers so that the devices appear to the operating system as direct-attached storage.

Enterprise Storage Options (continued)

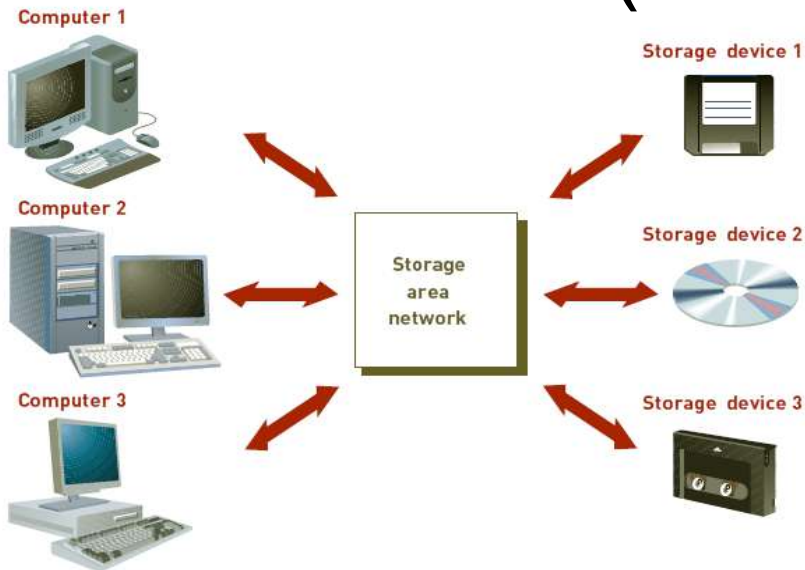
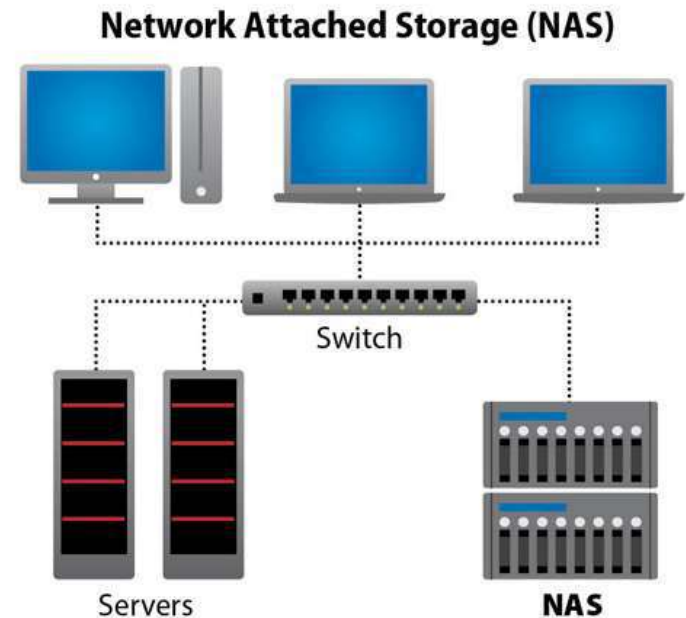


Figure (2.5): SAN and NAS Storage

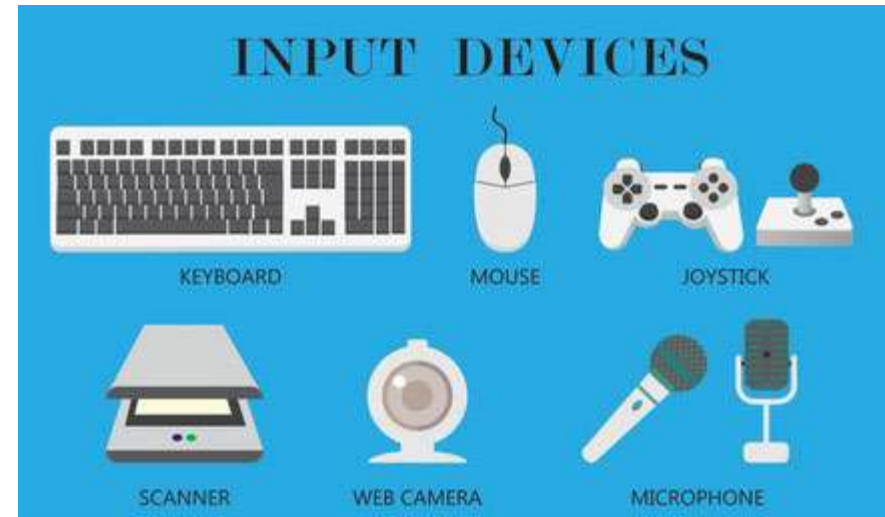


Input and Output Devices: The Gateway to Computer Systems

- Input and output devices
 - Gateways to the computer system
 - Part of a computer's user interface
- Data can be human-readable or machine-readable
 - Data entry: Converts human-readable data into machine-readable form
 - Data input: Transfers machine-readable data into the system

Input Devices

- Personal computer input devices
 - Keyboard
 - Mouse
 - Speech-recognition technology
 - Digital cameras
 - Scanning devices
 - Optical data readers
 - Magnetic stripe card
 - Point-of-sale devices (POS)
 - Automated teller machine (ATM) devices
 - Pen input devices
 - Touch-sensitive screens
 - Bar-code scanners
 - Radio frequency identification



Output Devices

- Display monitors (CRT, LCD, LED)
- Printers
- Plotters
- Speakers

SPEAKER



MONITOR



HEADPHONE



Output Devices of Computer

PLOTTER



PROJECTOR



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PRINTER



Computer System Types

- **Special-purpose computers**

- Used for limited applications by military and scientific research groups such as the CIA and NASA



- **General-purpose computers**

- Used for a wide variety of applications
- Handheld computer, desktop computer, portable computer, workstation, servers.





Introduction to Information Systems

Lecture Three

Software: Systems and Application Software



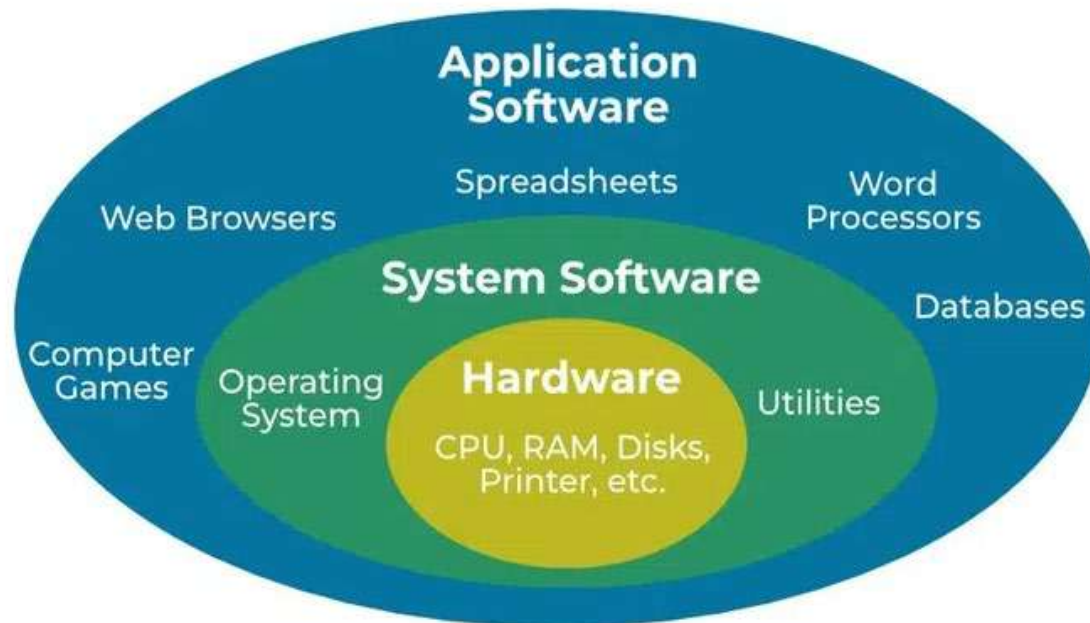
Dr. Mahmoud Shuker Mahmoud
mahmoud.shukur@mtu.edu.iq

Principles and Learning Objectives

- Systems and application software are critical in helping individuals and organizations achieve their goals
- Organizations use off-the-shelf application software for common business needs and proprietary application software to meet unique business needs and provide a competitive advantage
- The software industry continues to undergo constant change; users need to be aware of recent trends and issues to be effective in their business and personal life

An Overview of Software

- **Software** is a set of instructions, data or programs used to operate computers and execute specific tasks.
- Types of software:
 - Systems software
 - Application software,



Systems Software

- Controls the operations of computer hardware
- Supports the application programs' problem-solving capabilities
- Types of systems software:
 - Operating systems
 - Utility programs
 - Middleware

Operating Systems

Set of programs that controls computer hardware and acts as an interface with application programs

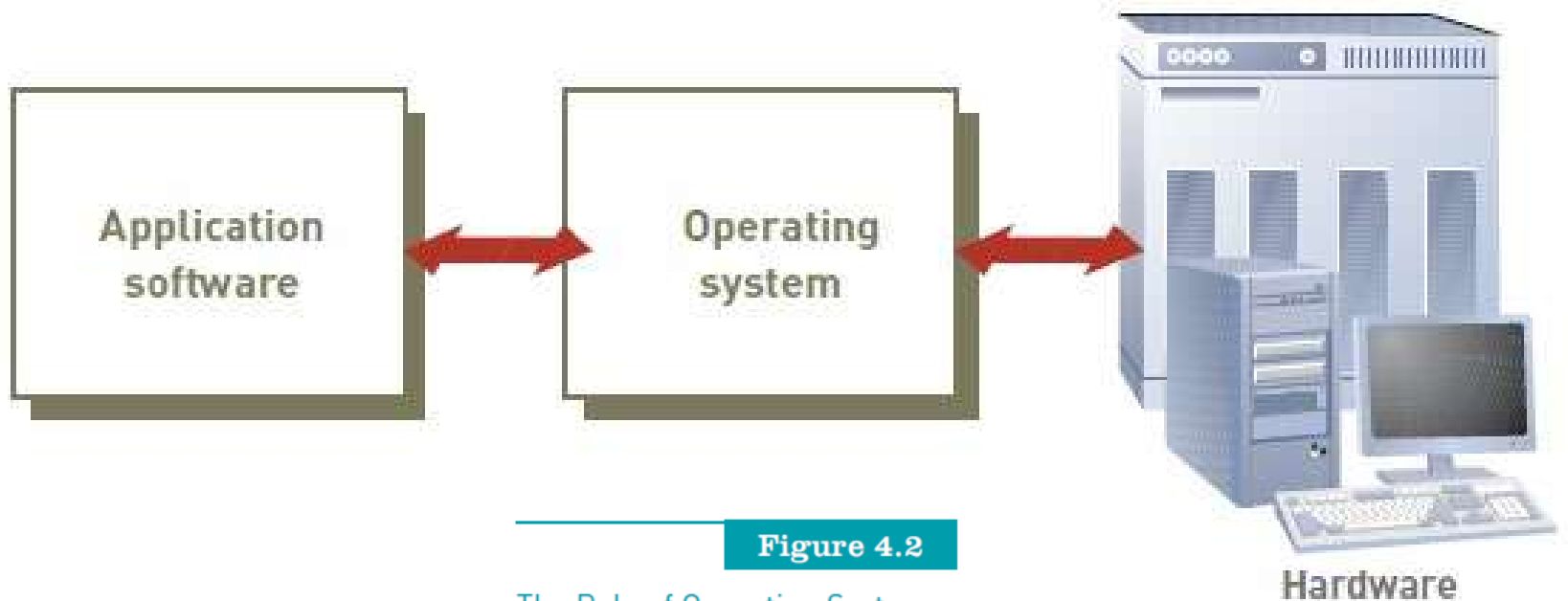


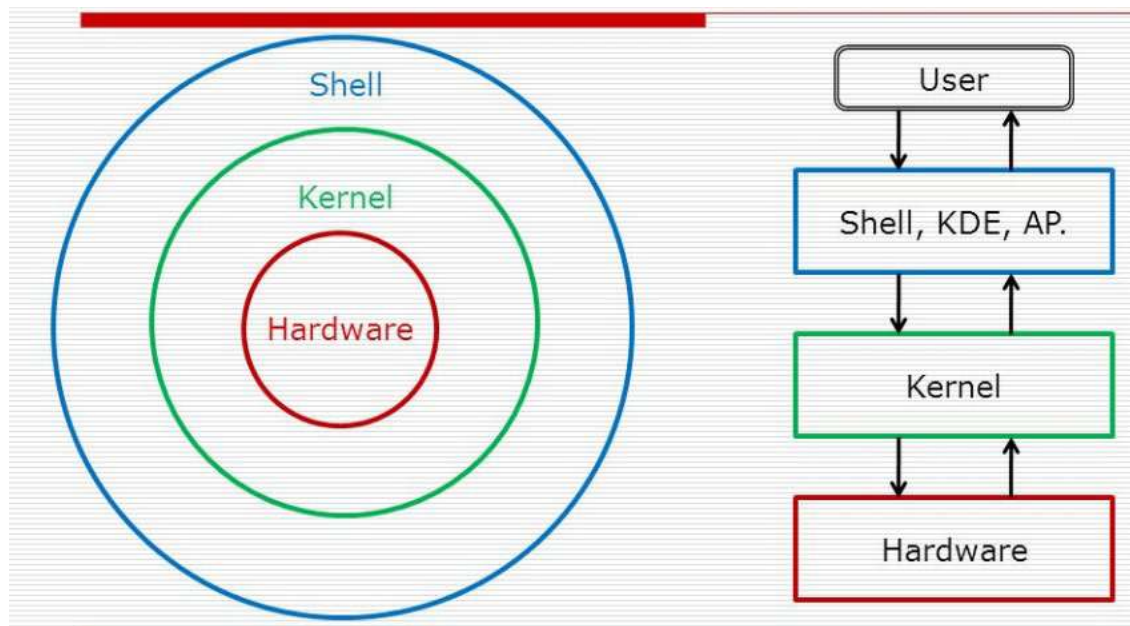
Figure 4.2

The Role of Operating Systems

The role of the operating system is to act as an interface between application software and hardware.

Operating Systems (continued)

- **Kernel:** Acts as a bridge between the applications and the processing done at the hardware level. It is responsible for accessing computer resources such as RAM and I/O devices.
- **Shell:** A computer program that exposes an operating system's services to a human user or other programs.



Operating Systems (continued)

- Activities performed by the operating system:
 - 1- Perform common computer hardware functions
 - 2- Provide a user interface and input/output management
 - 3- Provide a degree of hardware independence
 - 4- Manage system memory
 - 5- Manage processing tasks
 - 6- Provide networking capability
 - 7- Control access to system resources
 - 8- Manage files

Operating Systems (continued)

- **Common hardware functions:**
 - Get input from the keyboard or another input device
 - Retrieve data from disks
 - Store data on disks
 - Display information on a monitor or printer
- **User interface and input/output management:**
 - Command-based user interface: Requires that text commands be given to the computer to perform basic activities
 - Graphical user interface (GUI): Uses icons and menus displayed on the screen to send commands to the computer system.

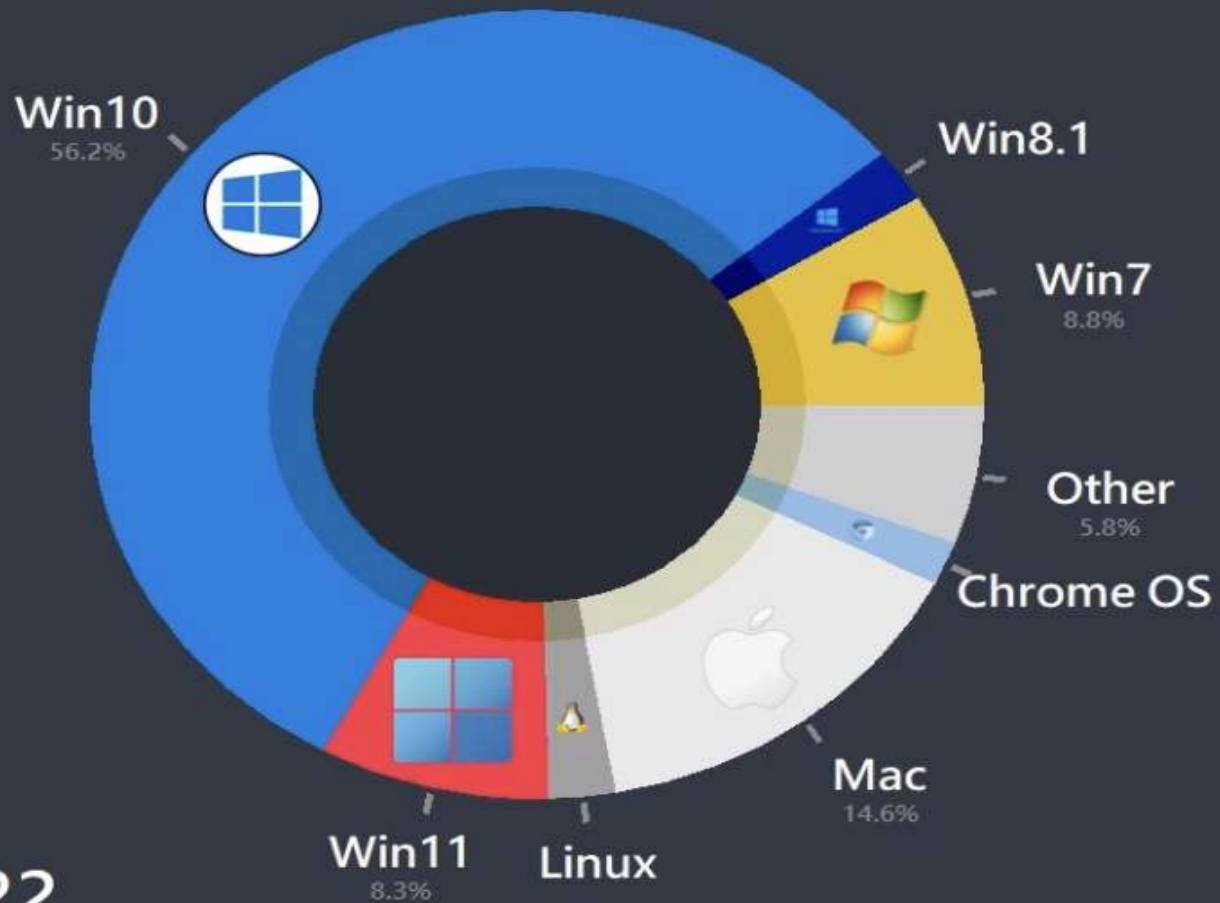
Operating Systems (continued)

- **Hardware independence:**
 - Application program interface (API): Allows applications to make use of the operating system
- **Memory management:**
 - Allows a computer to execute program instructions effectively and to speed up the processing
- **Processing tasks:**
 - Multitasking: More than one program can run at the same time
 - Time-sharing: Allows more than one person to use a computer system at the same time

Operating Systems (continued)

- **Networking capability:**
 - Allows computers in a network to send and receive data and share computing resources
- **Access to system resources and security:**
 - Protection against unauthorized access
- **File management:**
 - Ensures that files in secondary storage are available when needed and that they are protected from access by unauthorized users

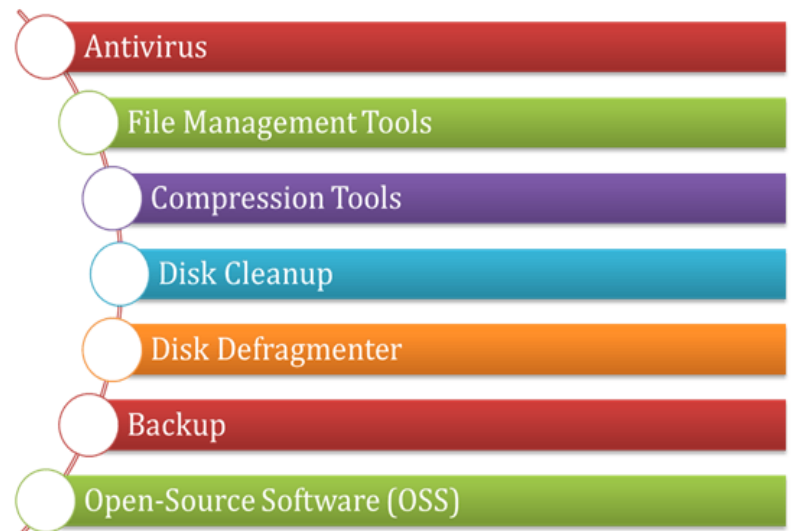
Current Operating Systems



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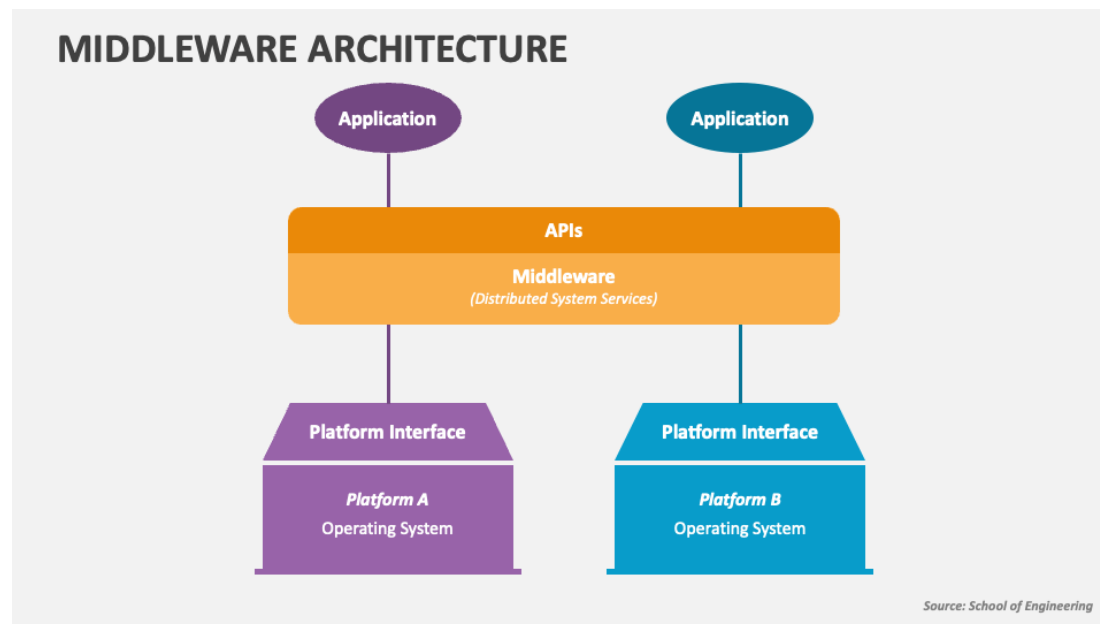
Utility Programs

- Help to perform maintenance or correct problems with a computer system.
- Common types of utilities:
 - Hardware utilities
 - Security utilities
 - File-compression utilities
 - Spam-filtering utilities
 - Network and Internet utilities
 - Server and mainframe utilities



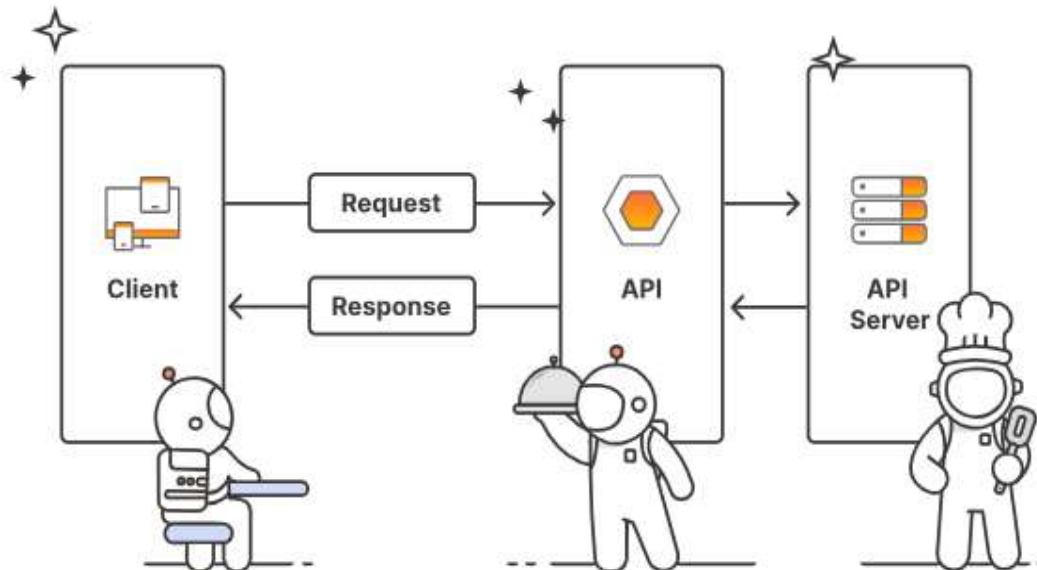
Middleware

- Software that allows different systems to communicate and exchange data
- Can also be used as an interface between the Internet and older legacy systems
- Uses modular application services to allow users to interact with systems and systems to interact with each other



Middleware

- **API:** which stands for Application Programming Interface, is a set of protocols that enable different software components to communicate and transfer data.
- Developers use APIs to bridge the gaps between small, discrete chunks of code to create applications that are powerful, resilient, secure, and able to meet user needs.
- Even though you can't see them, APIs are everywhere working continuously in the background to power the digital experiences essential to our modern lives.



Application Software

- Application programs:
 - Interact with systems software
 - Help you perform common tasks, such as:
 - Creating and formatting text documents
 - Performing calculations
 - Managing information



Overview of Application Software

- **Proprietary software:** A program for a specific application, usually developed and owned by a single company
- **Off-the-shelf software:** An existing software program that is purchased
- **Application service provider (ASP):** Company that can provide software, support, and computer hardware on which to run the software from the user's facilities over a network

Overview of Application Software (continued)

- **Software as a service (SaaS):**
 - Allows businesses to subscribe to Web-delivered business application software by paying a monthly service charge or a per-use fee
 - Can reduce expenses by sharing its running applications among many businesses
- **Cloud computing Platform:**
 - Use of computing resources on the Internet (the cloud) rather than on local computers

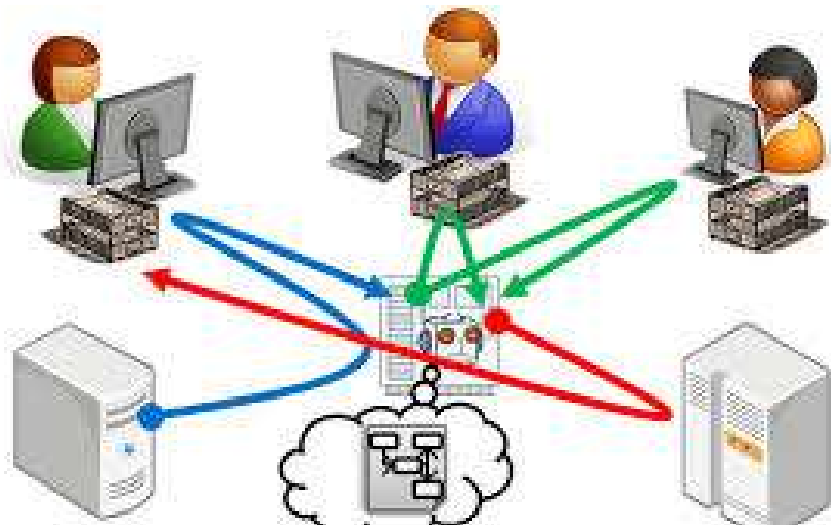
Personal Application Software

- Word processing
- Spreadsheet analysis
- Database applications
- Presentation graphics program
- Personal information managers
- Software suites and integrated software packages
- Other personal application software



Workgroup Application Software

- Support teamwork, whether people are in the same location or dispersed around the world
- **Groupware:** Software that helps groups of people work together more effectively



Programming Languages

- Sets of keywords, symbols, and a system of rules for constructing statements by which humans can communicate instructions to be executed by a computer.
- Program code: A set of instructions that signal the CPU to perform operations.
- Syntax: Set of rules associated with a programming language

The Evolution of Programming Languages

Generation	Language	Approximate Development Date	Sample Statement or Action
First	Machine language	1940s	00010101
Second	Assembly language	1950s	MVC
Third	High-level language	1960s	READ SALES
Fourth	Query and database languages	1970s	PRINT EMPLOYEE NUMBER IF GROSS PAY> 1000
Beyond Fourth	Natural and intelligent languages	1980s	IF gross pay is greater than 40, THEN pay the employee overtime pay

Table 4.10

The Evolution of Programming Languages

Visual, Object-Oriented, and Artificial Intelligence Languages

- **Visual programming:** Uses a graphical or “visual” interface combined with text-based commands
- **Object-oriented programming languages:** Programming languages that are based on objects
- **Fifth-generation languages (5GLs):** Used to create artificial intelligence or expert systems applications
- **Compiler:** Converts the programmer’s source code into machine-language instructions

Visual, Object-Oriented, and Artificial Intelligence Languages (continued)

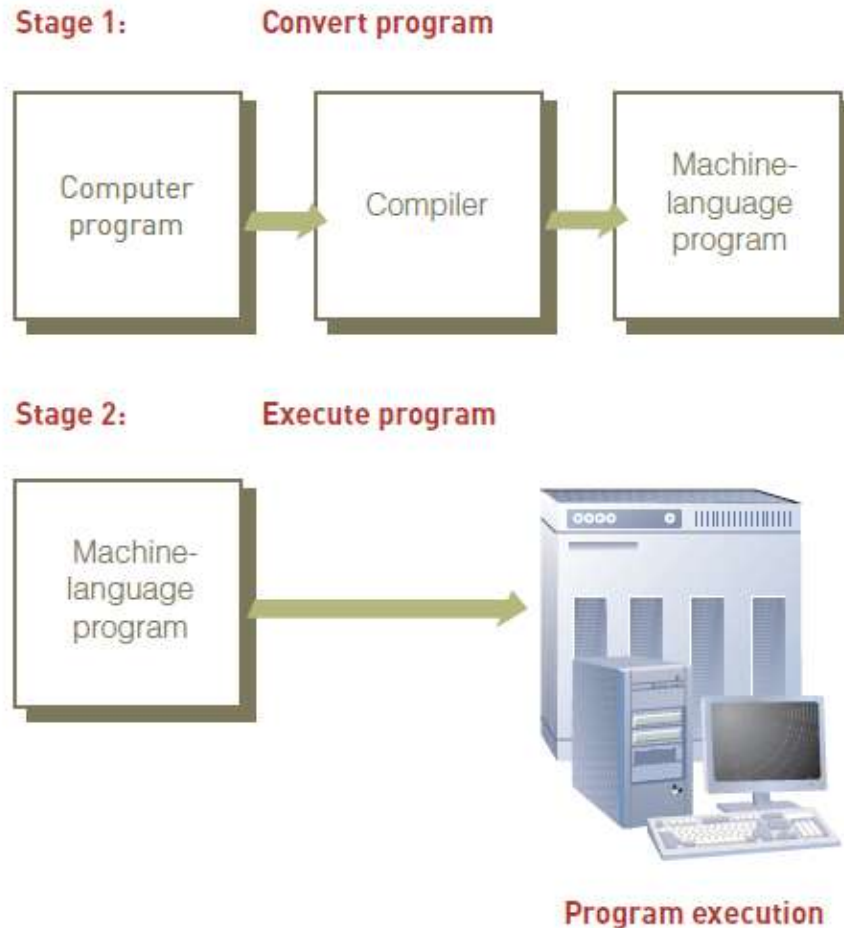


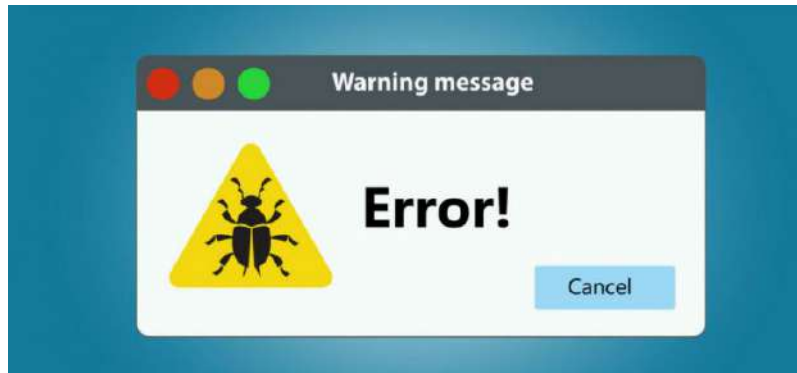
Figure 4.18

How a Compiler Works

A compiler translates a complete program into a complete set of binary instructions (Stage 1). After this is done, the CPU can execute the converted program in its entirety (Stage 2).

Software Issues and Trends

- **Software bug:** A defect in a program that keeps it from performing as it should
- Some tips for reducing the impact of software bugs:
 - Register all software
 - Check readme files for workarounds
 - Access the support area of the manufacturer's Web site for patches
 - Install the latest software updates



Copyrights and Licenses

- Most software products are protected by law using copyright or licensing provisions:
 - In some cases, you are given unlimited use of software on one or two computers
 - In other cases, you pay for your usage; if you use the software more, you pay more
- Some software now requires that you *register* or *activate* it before it can be fully used

Copyrights and Licenses (continued)

License	Description
Single-user license	Permits you to install the software on one computer, or sometimes two computers, used by one person.
Multiuser license	Specifies the number of users allowed to use the software, and the number that can be installed on each user's computer. For example, a 20-user license can be installed on 20 computers for 20 users.
Concurrent-user license	Designed for network-distributed software, this license allows any number of users to use the software, but only a specific number of users to use it at the same time.
Site license	Permits the software to be used anywhere on a particular site, such as a college campus, by everyone on the site.

Table 4.11

Software Licenses

Freeware and Open-Source Software

- **Freeware:** Software that is made available to the public for free
- **Open-source software:** Distributed, typically for free, with the source code
- **General Public License (GPL)** grants you the right to:
 - Run the program for any purpose
 - Study how the program works and adapt it to your needs
 - Redistribute copies so you can help others
 - Improve the program and release improvements to the public

Software Upgrades

- Software companies revise their programs and sell new versions periodically
- Most software upgrades offer new features and capabilities
- Some users do not always get the most current software upgrades or versions unless it includes significant improvements or capabilities





Introduction to Information Systems

Lecture Four Telecommunications and Networks



Dr. Mahmoud Shuker Mahmoud
mahmoud.shukur@mtu.edu.iq

Principles and Learning Objectives

- Identify and describe the fundamental components of a telecommunications system.
- Networks are an essential component of an organization's information technology infrastructure
 - Identify the benefits of using a network
 - Describe the different types of network topologies.
 - Identify network categories.
 - Study LAN transmission methods and devices
 - Describe the guided and unguided transmission media types.
 - Study the OSI reference model and its layers.

DATA COMMUNICATIONS

The term telecommunication means communication at a distance. Data communications are the exchange of data between two devices via a transmission medium such as a wire cable or wireless.

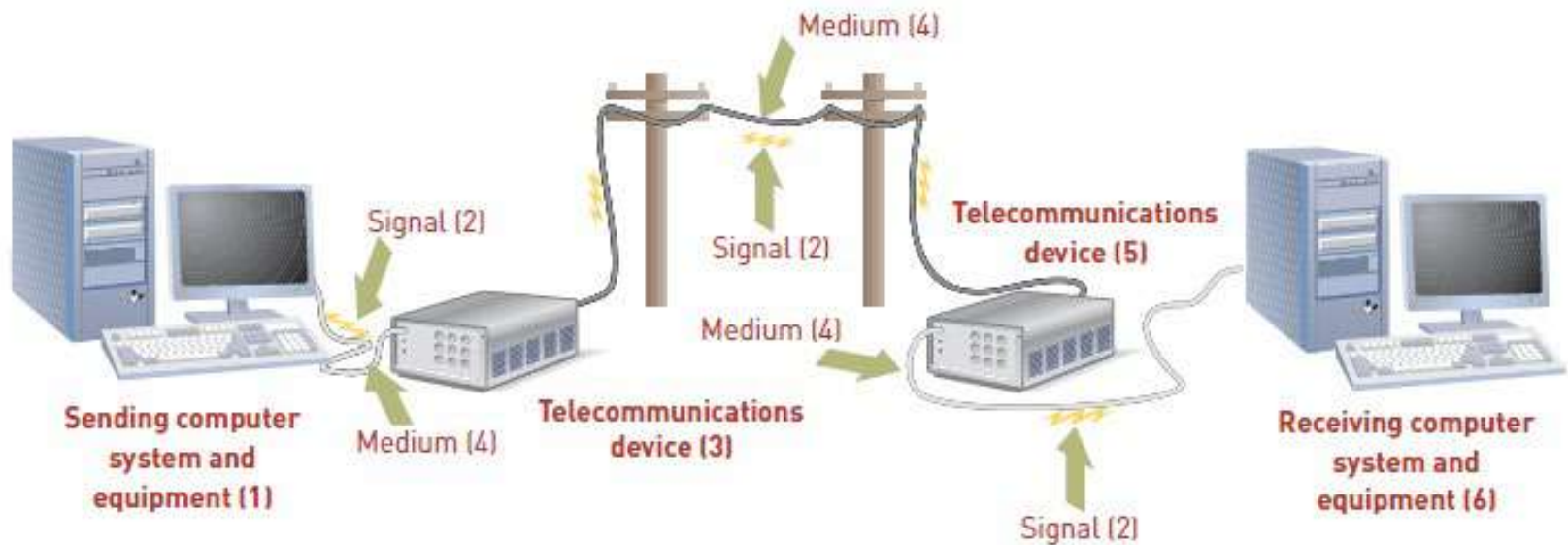


Figure (4.1) Elements of a Telecommunication System

Communication System Components

Five components of data communication systems (Networks)

- 1- Sender (Source)
- 2- Receiver (Destination)
- 3- Medium
- 4- Message
- 5- Protocol

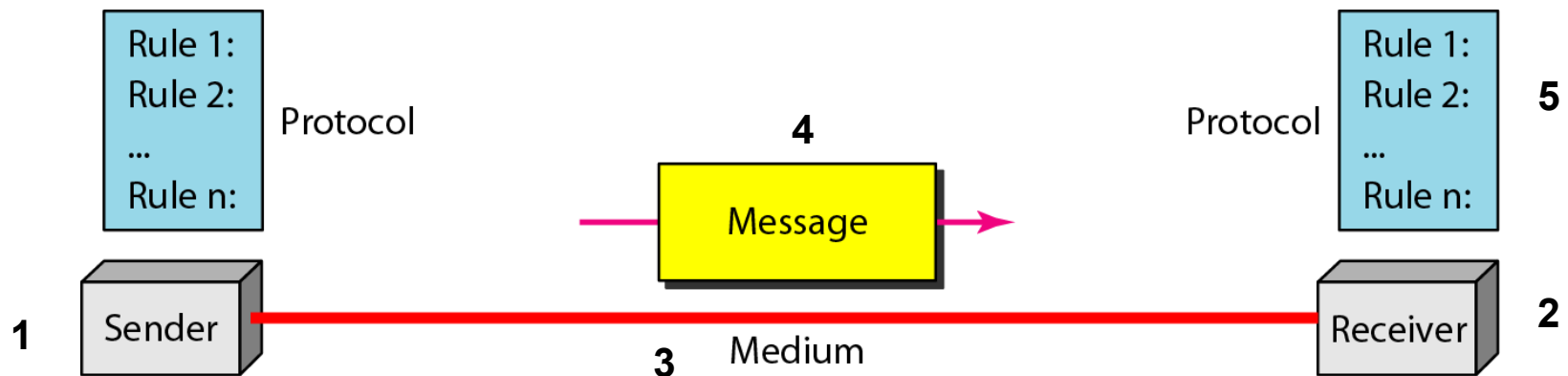


Figure (4.2) Communication System Components

Data flow in Communication Systems

- ❖ Simplex: Flow of data in one direction only.
- ❖ Half-duplex: Flow of data in two direction but at different time.
- ❖ Full-duplex: Flow of data in both direction at the same time.

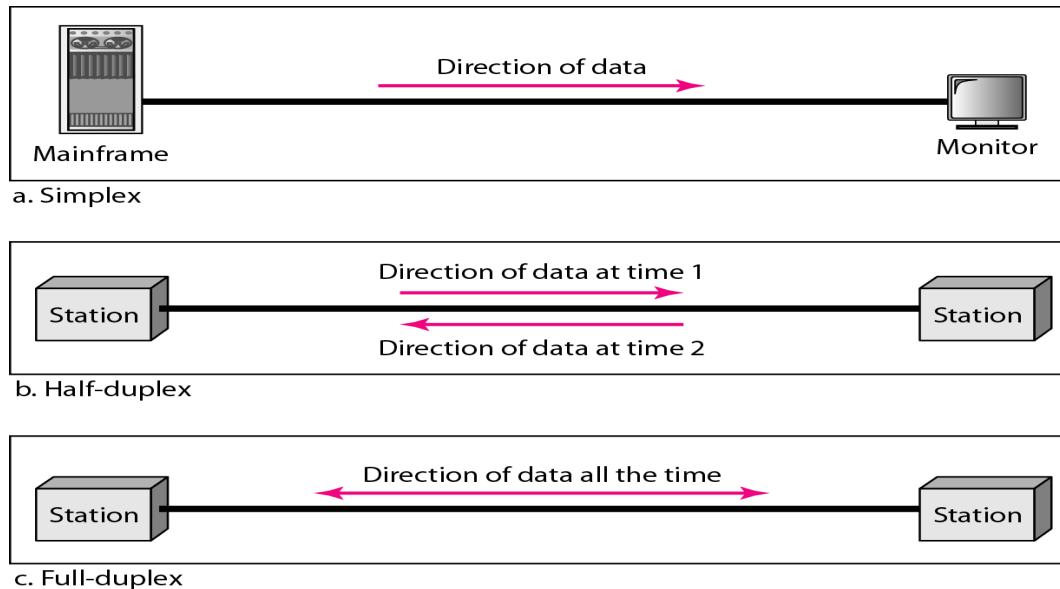
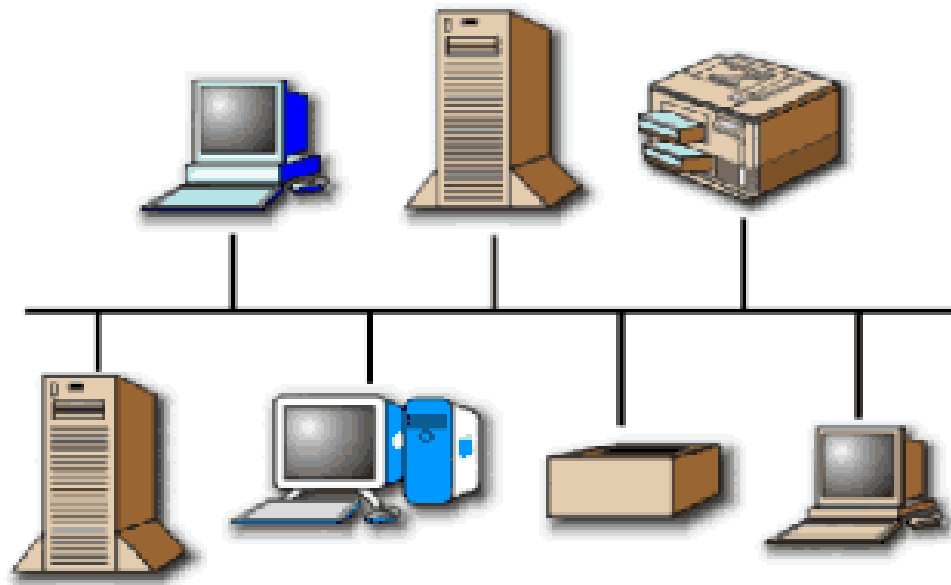


Figure (4.3) Data Flow Types

NETWORKS

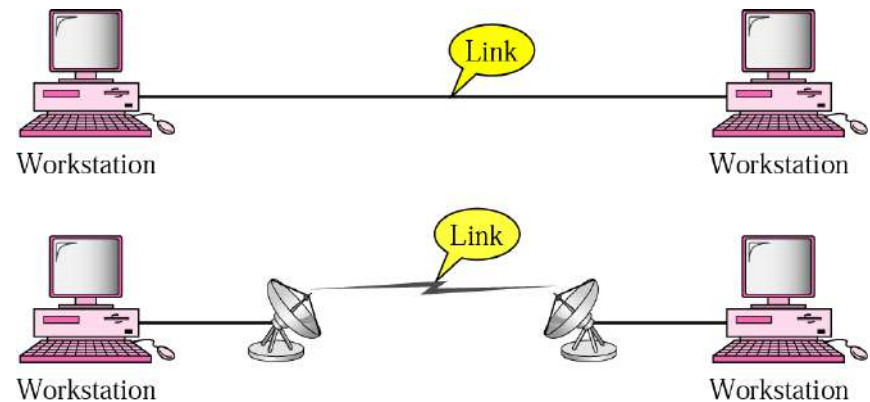
A network is a set of devices (nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and receiving data generated by other nodes on the network.



Types of connections

■ Point to point

Point-to-point communication means the channel is shared between two devices.



■ Multipoint

Multipoint Communication means the channel is shared among multiple devices or nodes.

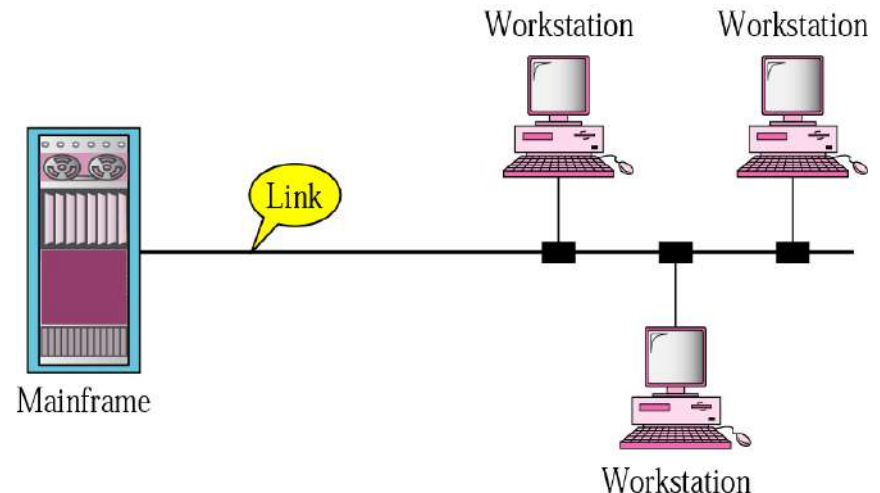


Figure (4.4) Types of Connections

Network Topology

Network topology refers to the manner in which the links and nodes of a network are arranged to relate to each other. Topologies are categorized into six types:

- 1- Mesh Topology.
- 2- Star Topology.
- 3- Bus Topology.
- 4- Ring Topology.
- 5- Tree Topology.
- 6- Hybrid Topology.

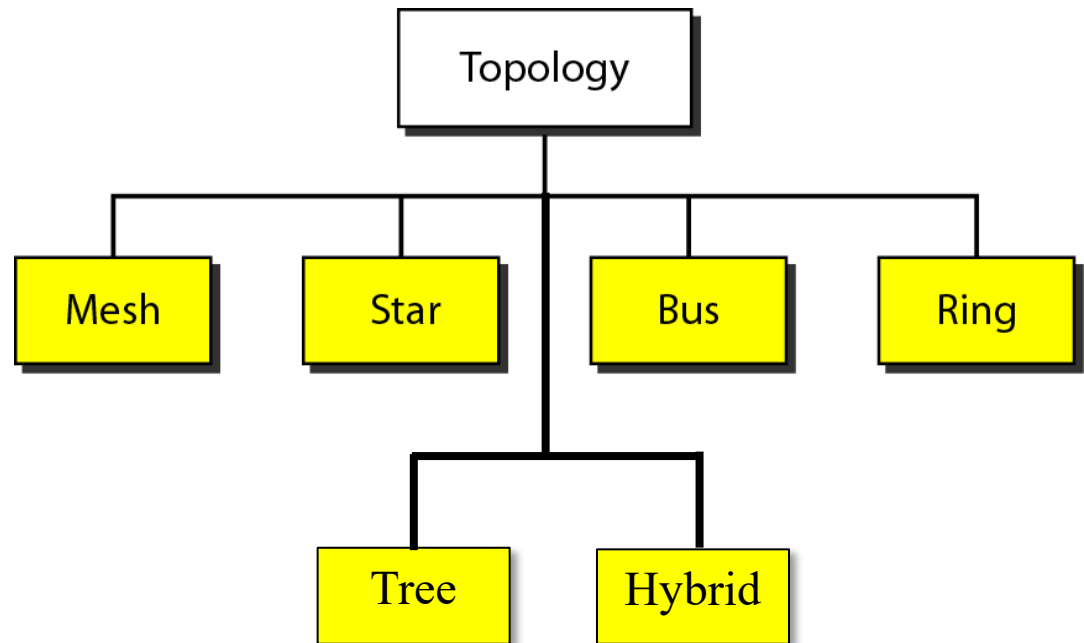
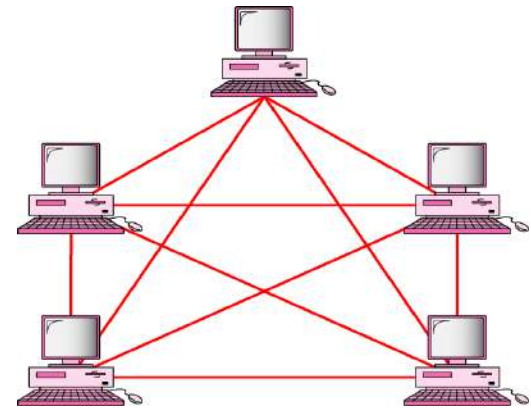


Figure (4.5) Categories of Network Topologies

MESH Topology

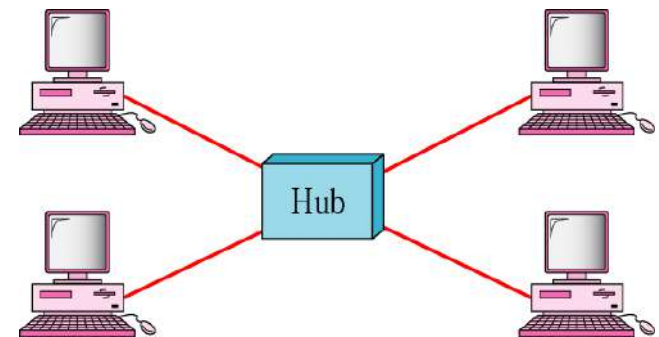
- Every device has a dedicated point-to-point link to every other device.

$n(n-1)/2$ physical duplex links



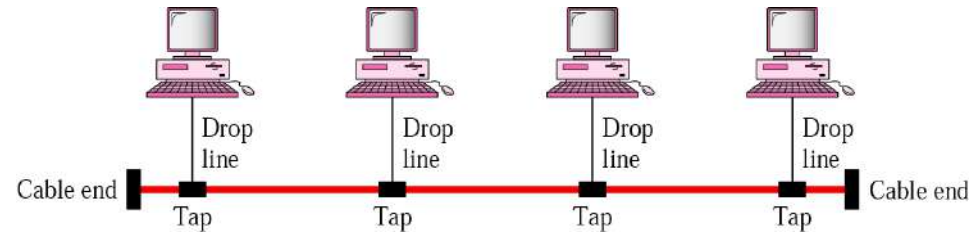
STAR Topology

- Each device has a dedicated point-to-point link only to a central controller, usually called a hub or a switch.
- No direct traffic and link between devices



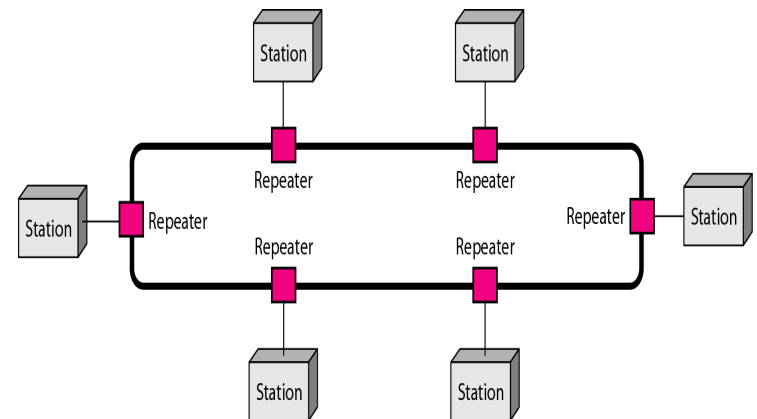
BUS Topology

- A multipoint topology
- All devices are linked through a backbone cable
- Nodes are connected to the bus cable by drop lines and taps.



RING Topology

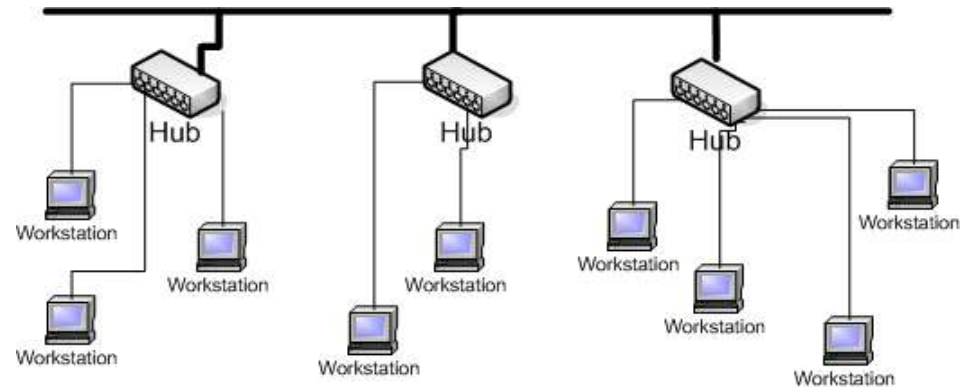
- Each device is dedicated point-to-point connection only with the two devices on either side of it
- A signal is passed along the ring in the direction, from device to device, until it reaches its destination
- Each device in the ring incorporates a repeater



Tree Topology

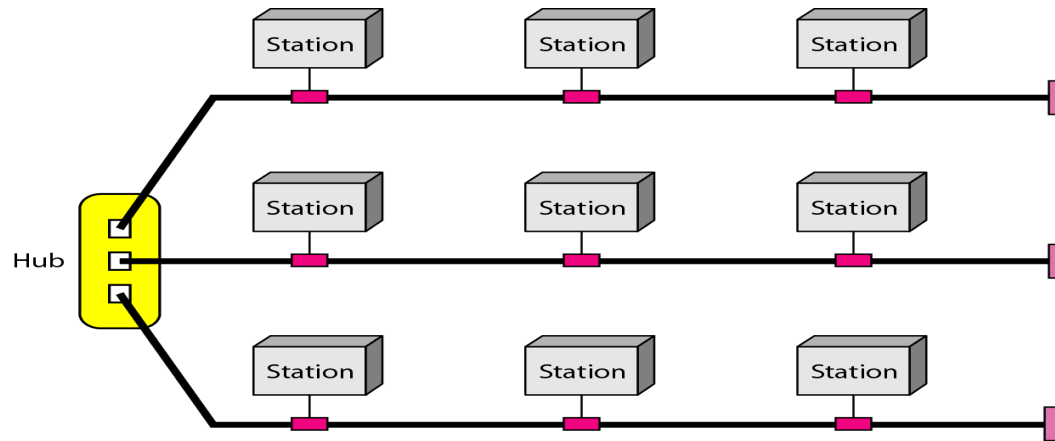
Tree topologies integrate multiple topologies together

Example: Tree topology integrates multiple star topologies together onto a bus



Hybrid Topology

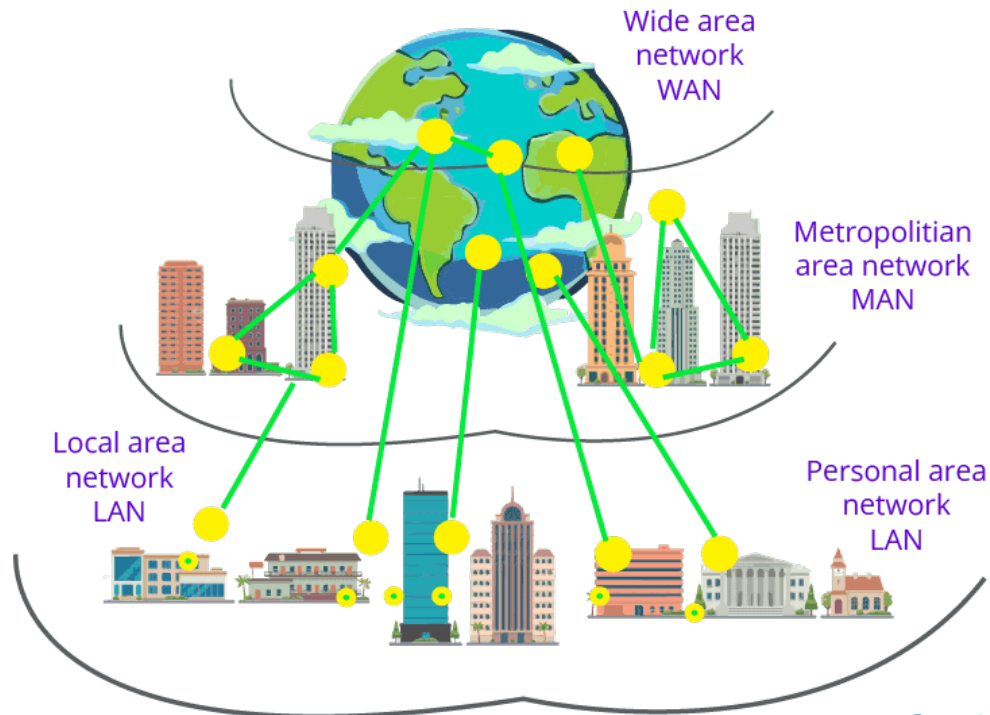
Hybrid topology integrates multiple topologies together. *For Example, integrating a star backbone with three bus networks*



Categories of Networks

Networks can be categorized according to their size into:

1. Personal Area Network (PAN)
2. Local Area Network (LAN)
3. Metropolitan Area Network (MAN)
4. Wide Area Network (WAN)



NETWORKS EXPLAINED

PAN

A PAN (Personal Area Network) is a small network based on an individual and can connect to multiple devices. They are only used in one building.



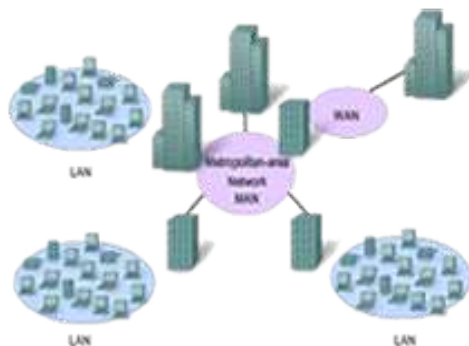
LAN

A LAN (Local Area Network) is a bigger network than the PAN, however still small. It can cover larger buildings than the PAN, such as an office building or apartment block, but cannot be used outside the building.



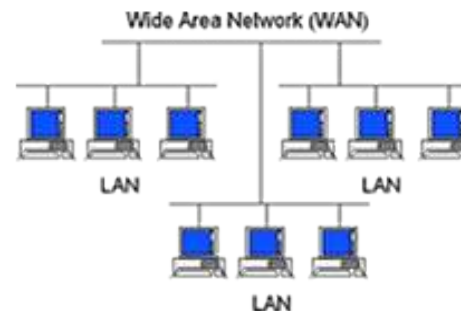
MAN

A MAN (Metropolitan Area Network) is much larger than the PAN or LAN networks, as it can cover entire towns or cities, using several LANs to create a bigger network.



WAN

A WAN (Wide Area Network) is the largest network in the world. It uses lots of different LANs and MANs to create huge networks that can cover a whole country or even the whole world. The World Wide Web (WWW) is an example of a public WAN.



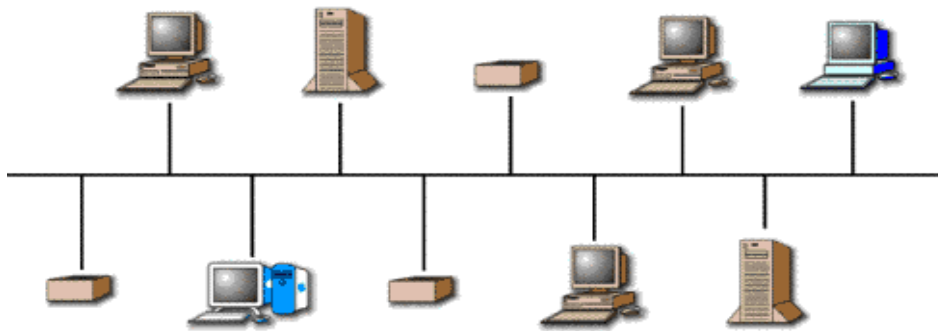
LAN Transmission Methods

LAN transmission methods fall into 3 main categories:

- **Unicast transmission**
- **Multicast transmission**
- **Broadcast transmission**

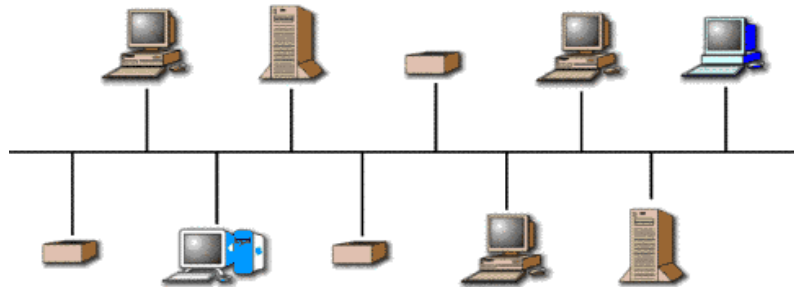
Unicast Transmission

In unicast transmissions, a single data packet is sent from a source to a single destination on the network.



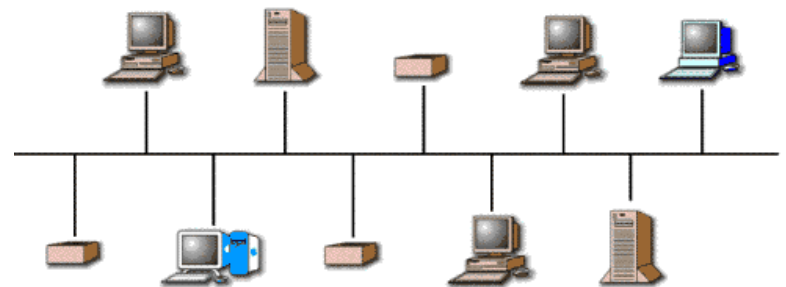
Multicast Transmission

In multicast transmissions, a single data packet is copied and sent to specific destinations on the network.



Broadcast Transmission

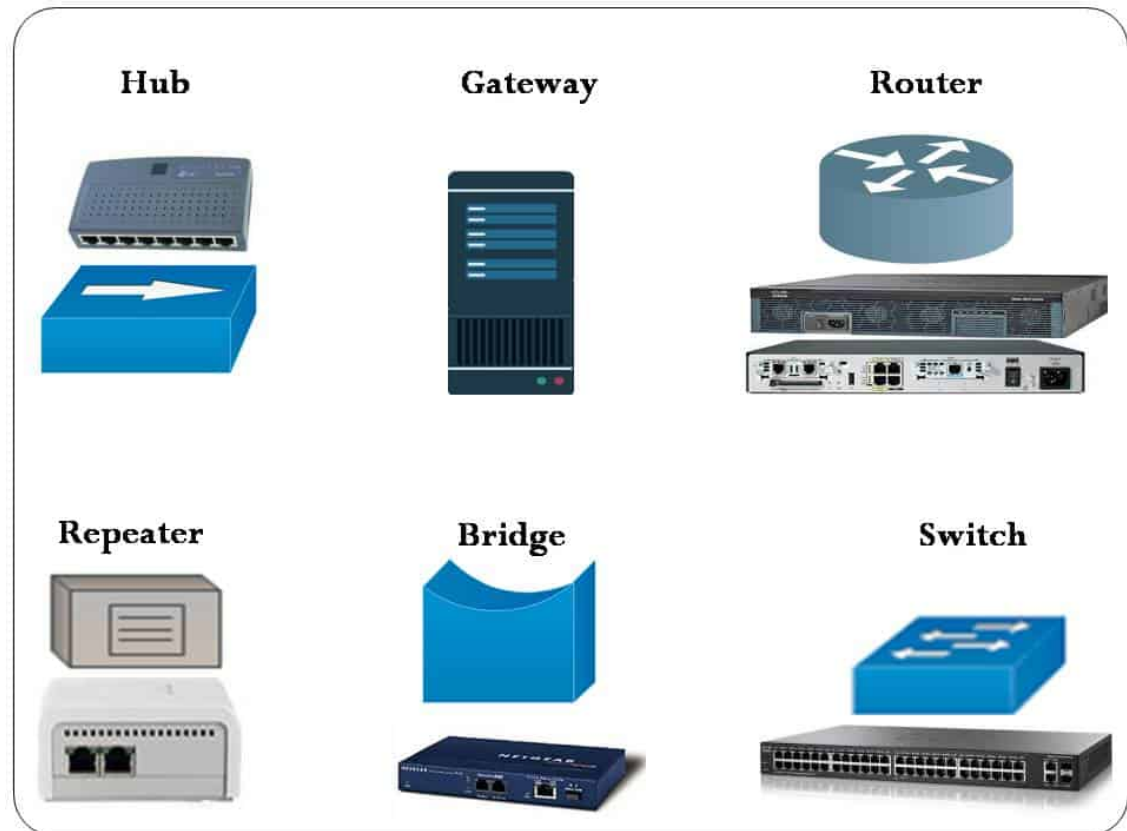
In broadcast transmissions, a single data packet is copied and sent to all destinations on the network.



LAN Infrastructure Devices

There are numerous devices associated with data information flow across a LAN. When adjoined, they create the infrastructure of a functional LAN. These devices include:

- Repeaters
- Bridges
- Hubs
- Switches
- Routers





Bridge

A network bridge acts as an interconnection between two LANs, creating a single network from separate LANs.



Gateway

A gateway connects discrete networks and translates packet data so it can travel between the systems.



Hub

A hub joins multiple devices on the same LAN, broadcasting messages to all ports without examining frames.



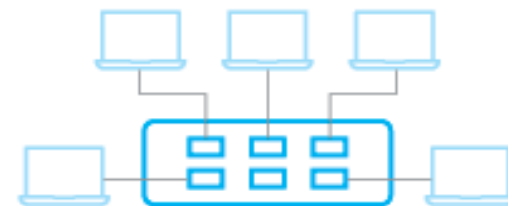
Repeater

A repeater strengthens a signal and retransmits it along to its destination.



Router

A router directs data requests from one network to another, using a packet's IP address to forward it to its destination.



Switch

A network switch forwards data to its proper destination, examining a packet's MAC address info to determine the intended device.

Network Cabling

Network Media can be classified into two distinct types:

1- Guided transmission media.

2- Unguided transmission media.

Guided transmission media is available in many types:

- **Twisted-pair wire:** Classified by category: category 2, 3, 5, 5E, 6, and 7.
- **Coaxial cable:** Offers cleaner and crisper data transmission (less noise) than twisted-pair wire.
- **Fiber-optic cable:** Transmits signals with light beams

The earliest LANs used coaxial cables. Over time, the twisted pair cables used in telephone systems were improved to carry higher frequencies and support LAN traffic. More recently, fiber optic cables have emerged as a high-speed cabling option.

Guided Transmission Media Characteristics

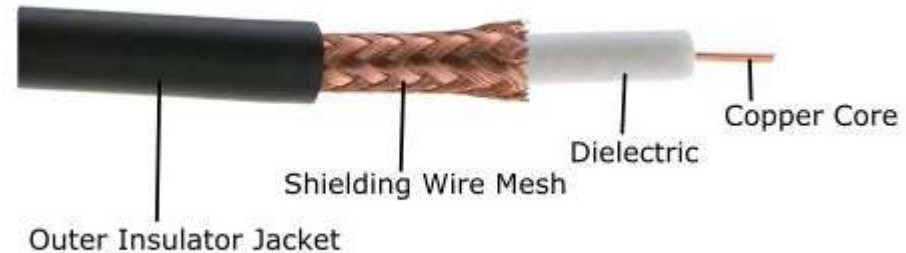
Media Type	Description	Advantages	Disadvantages
Twisted-pair wire	Twisted pairs of copper wire, shielded or unshielded	Used for telephone service; widely available	Transmission speed and distance limitations
Coaxial cable	Inner conductor wire surrounded by insulation	Cleaner and faster data transmission than twisted-pair wire	More expensive than twisted-pair wire
Fiber-optic cable	Many extremely thin strands of glass bound together in a sheathing; uses light beams to transmit signals	Diameter of cable is much smaller than coaxial; less distortion of signal; capable of high transmission rates	Expensive to purchase and install
Broadband over power lines	Data is transmitted over standard high-voltage power lines	Can provide Internet service to rural areas where cable and phone service may be nonexistent	Can be expensive and may interfere with ham radios and police and fire communications

Coaxial Cables

A coaxial cable consists of:

- a single copper conductor
- a layer of shielding with a ground wire
- an outer jacket

Coaxial cables are sometimes used for bus topologies, but many LAN products are dropping support of coaxial cable connectivity.



The Ethernet LAN protocol was initially developed to operate over coaxial cables. It is no longer recommended but still used in some small LANs.

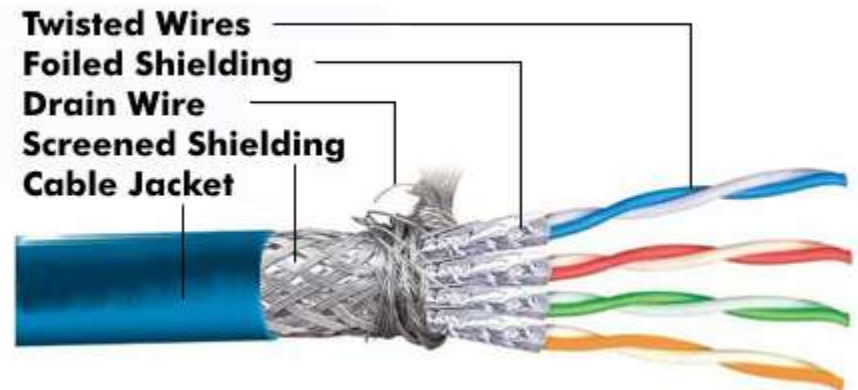
Unshielded Twisted Pair



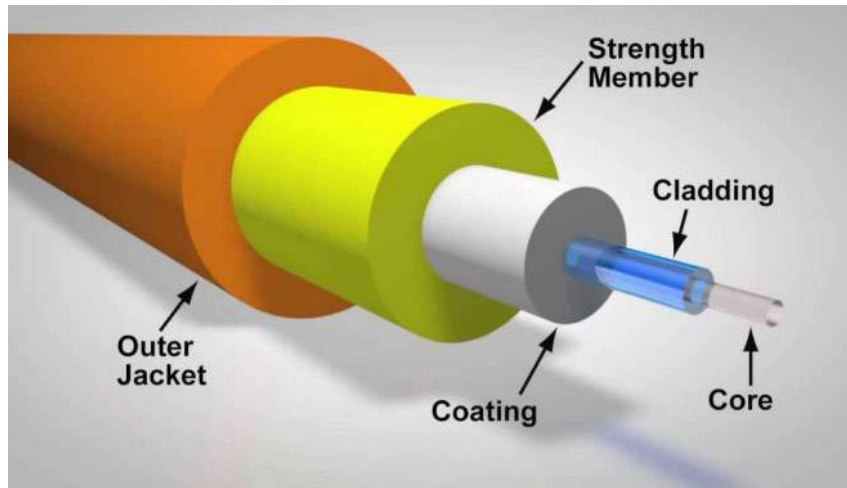
Unshielded twisted pair (UTP) cable is used for both LANs and telephone systems. UTP cables are composed of four color-coded pairs of copper conductors twisted around each other. An outer jacket provides protection and keeps the pairs in alignment. UTP cable connects to devices via 8 pin modular connectors called RJ-45 plugs. All LAN protocols can operate over UTP. Most modern LAN devices are equipped with RJ-45 jacks.

Shielded Twisted Pair

STP cable is also used for Data Networks. Its shielding allows greater tolerances for protection from EMI interference, such as from fluorescent light fixtures and electric motors.



Fiber Optic Cable



Fiber Optic cables are the latest development in cabling technology. They are constructed from optical glass. There is a central glass filament, called the core, and surrounding layers of cladding, buffer coatings, strengthening materials, and an outer jacket.

Information is transmitted by wavelengths of light. This is accomplished through devices that convert electrical signals into rapid pulses of either **LED** or Laser light.

Fiber optic cables offer several advantages, including:

- high bandwidth capacity (many gigabits per second).
- longer distances between devices (from 2 to over 60 kilometers).
- immunity to electromagnetic interferences

Fiber optic cables are widely used in WANs for both voice and data communications. The primary barrier to their widespread use in LANs is the cost of electronics.

Wireless Communication

Unguided transmission media involves the broadcast of communications in one of three frequency ranges (Radio, Microwave, and infrared frequencies)

Technology	Description	Advantages	Disadvantages
Radio frequency range	Operates in the 3KHz–300 MHz range	Supports mobile users; costs are dropping	Signal highly susceptible to interception
Microwave—terrestrial and satellite frequency range	High-frequency radio signal (300 MHz–300 GHz) sent through atmosphere and space (often involves communications satellites)	Avoids cost and effort to lay cable or wires; capable of high-speed transmission	Must have unobstructed line of sight between sender and receiver; signal highly susceptible to interception
Infrared frequency range	Signals in the 300 GHz–400 THz frequency range sent through air as light waves	Lets you move, remove, and install devices without expensive wiring	Must have unobstructed line of sight between sender and receiver; transmission effective only for short distances

Table 6.2

Frequency Ranges Used for
Wireless Communications

Short Range Wireless Options

- **Near field communication (NFC)**
 - Short-range wireless connectivity technology designed for cell phones and credit cards
- **Bluetooth**
 - Wireless communications specification that describes how cell phones, computers, personal digital assistants, etc., can be interconnected
- **Ultra wideband (UWB)**
 - Short-range communications that employ extremely short electromagnetic pulses lasting just 50 to 1,000 picoseconds

Short Range Wireless Options (continued)

- **Infrared transmission**
 - Sends signals at a frequency of 300 GHz and above
- **Zigbee**
 - Form of wireless communications frequently used in security systems and heating and cooling control systems

Medium-Range Wireless Options (continued)

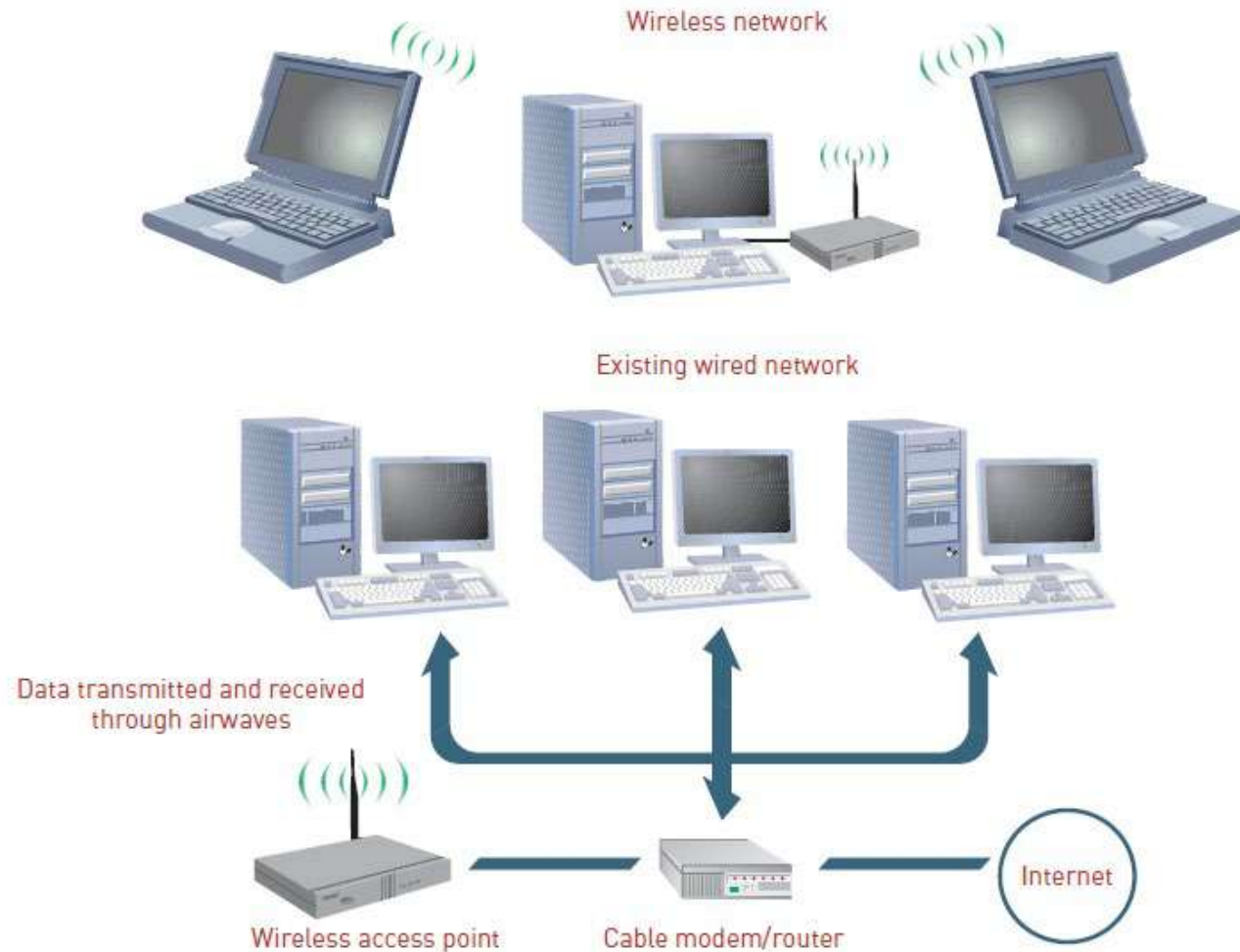
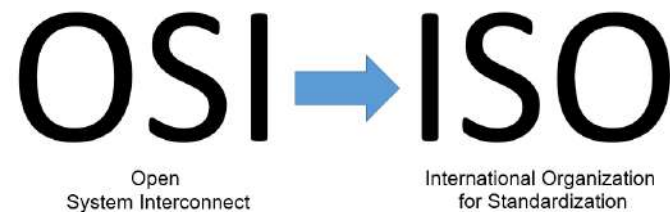


Figure 6.3
Wi-Fi Network

Network Model Overview

For a computer to send information to another computer and for that computer to receive and understand the information, a set of rules or standards must exist for this communication process. These standards ensure that varying devices and products can communicate with each other over any network. This set of standards is called a model.

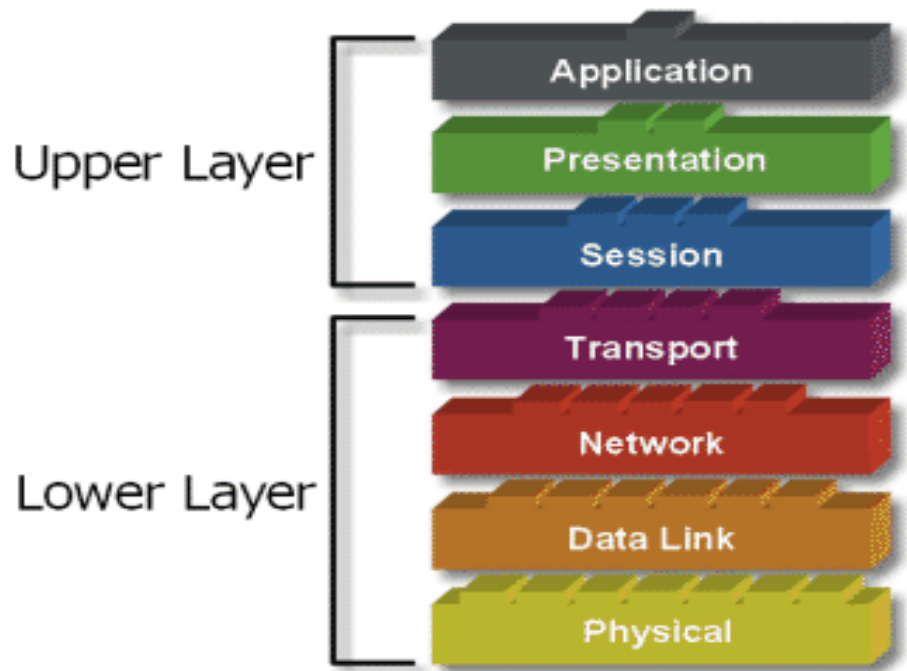
The most famous model in Computer Networks is called the OSI (Open System Interconnection) Model which is developed by ISO (international System Organization).



1984

OSI Network Model

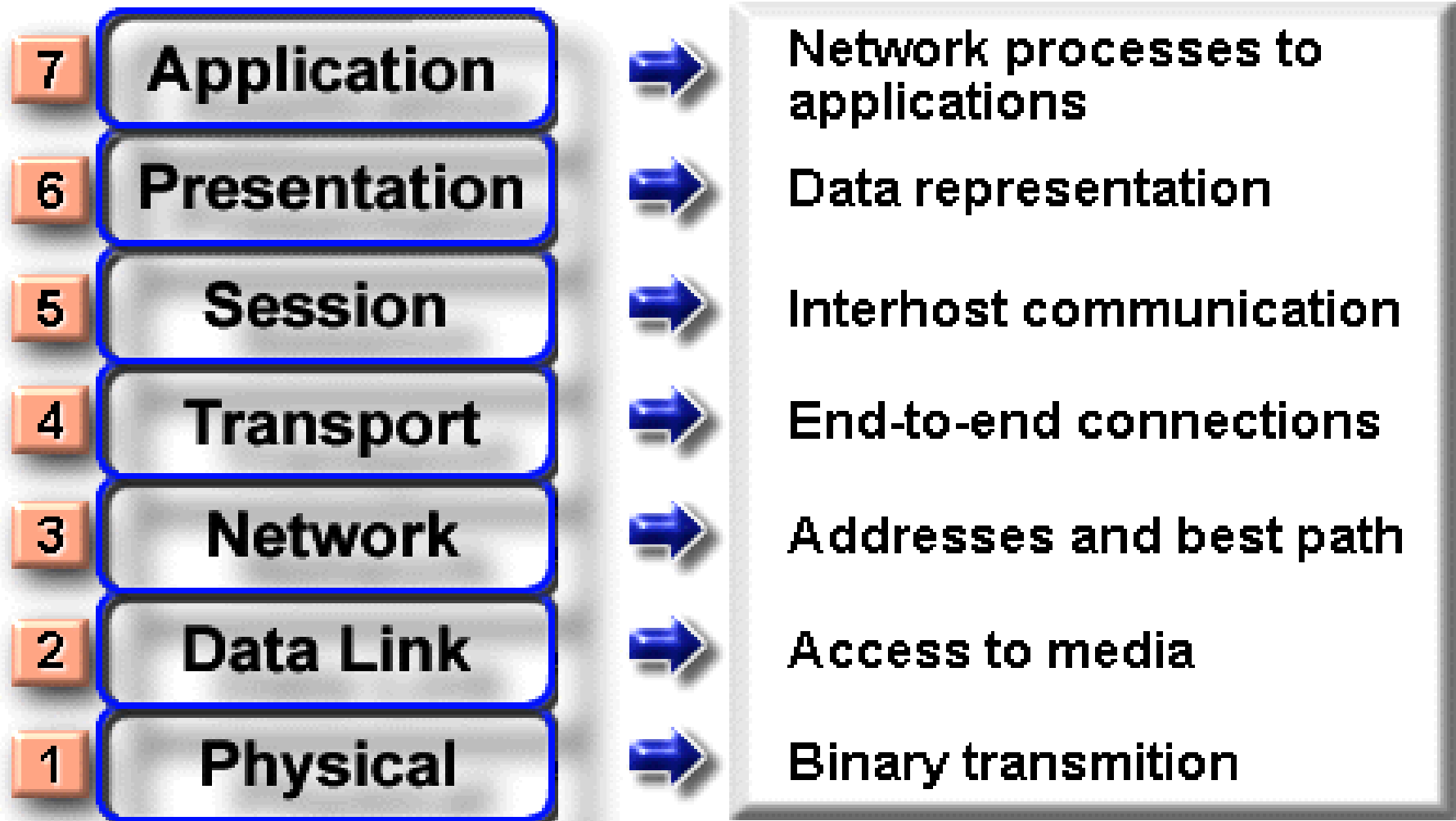
There are 7 layers in the OSI model. Each layer is responsible for a particular aspect of data communication. For example, one layer may be responsible for establishing connections between devices, while another layer may be responsible for error checking during transfer.



The layers of the OSI model are divided into two groups: **the upper layer** and lower layer. The upper layers focus on user applications and how files are represented on the computers prior to transport. For the most part, network engineers are more concerned with the **lower layers**. It's the lower layers that concentrate on how the communication across a network actually occurs.

- ▶ **ALL People Seem to Need Data Processing (Layer 7 to 1)**
- ▶ **Please Do Not Take Sausage Pizzas Away (Layer 1 to 7)**

Layer Functions



Securing Data Transmission

- **Encryption:** Converting an original message into a form that can only be understood by the intended receiver
- **Encryption key:** Variable value that is applied (using an algorithm) to a set of unencrypted text to produce encrypted text or to decrypt encrypted text

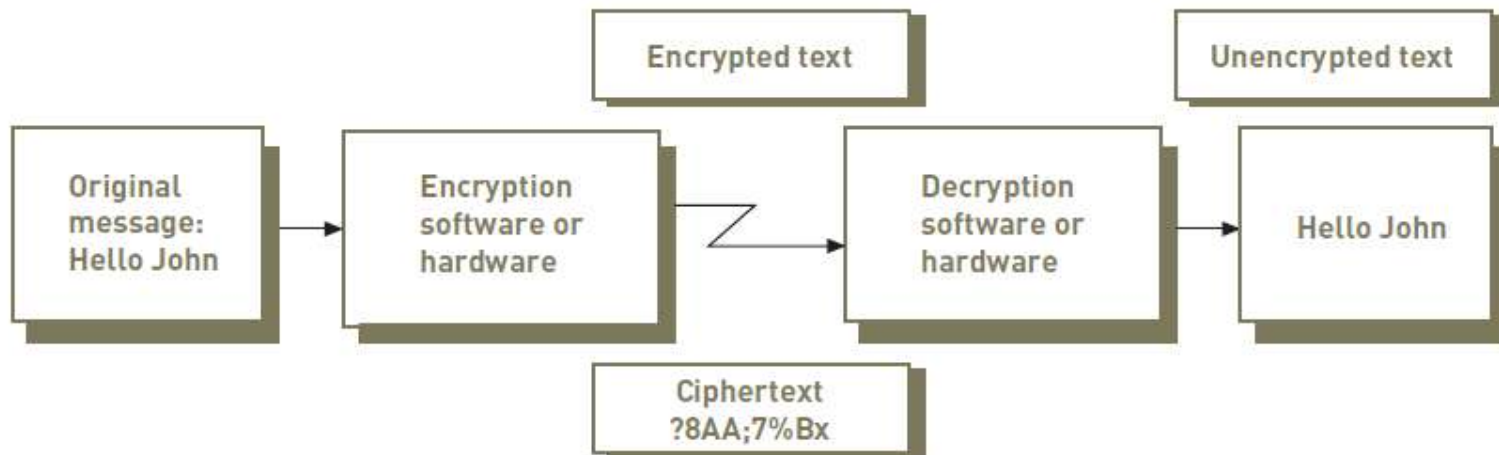


Figure 6.12

Encryption Process



Introduction to Information Systems

Lecture Five Networks Security



Dr. Mahmoud Shuker Mahmoud
mahmoud.shukur@mtu.edu.iq

Principles and Learning Objectives

- Identify and describe the fundamental concepts of network security and cybersecurity.
- Network security involves studying the following
 - Identify the basic terms threats, attacks, and vulnerability
 - Describe the different types of network threats, attacks, and vulnerabilities.
 - Identify the different types of threats.
 - Identify the different types of attacks.
 - Study the different malware types and distinctions.
 - Describe the role of social engineering in cybersecurity.
 - Explain the meaning of information security and the CIA Triad.

Threats, Vulnerabilities, and Attacks

- When discussing **network security**, the three common terms used are:
- **Threat** is any malicious occurrence that could harm an asset. In other words, a threat is any bad thing that can happen to your assets.
- **Vulnerability** is a weakness that makes a threat possible. This may be because of poor design, configuration mistakes, or inappropriate and insecure coding techniques.
- **Attack** is an action that exploits a vulnerability. Examples of attacks include sending malicious input to an application or flooding a network in an attempt to deny service.

What are the differences between Attack and Threat?

- **The main difference** between a threat and an attack is that a threat can be **intentional** or **unintentional**, whereas an attack is **intentional**.
- Threat is a circumstance that has the potential to cause loss or damage, whereas an attack is an attempt to cause damage.
- A threat to the information system doesn't mean the information was altered or damaged. Still, an attack on the information system means there might be a chance to alter, damage, or obtain information when the attack is successful.

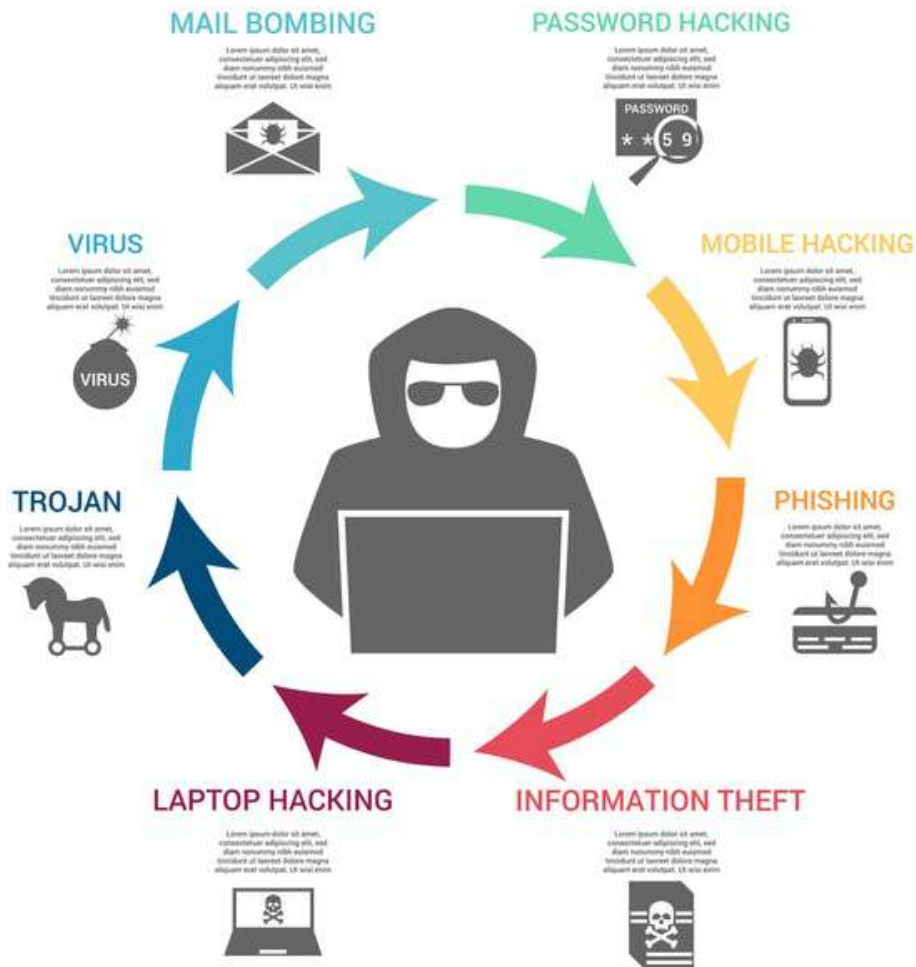
Vulnerability

- The vulnerabilities could be
 - ❑ Technological weaknesses
 - ❑ Configuration weaknesses
 - ❑ Security policy weaknesses
- Any discovered vulnerability must be **addressed to mitigate** any threat that could take advantage of the vulnerability.
- It is difficult to test an application thoroughly for all possible intrusions.
- **In summary**, vulnerability is a weakness that is inherent in every network and device. This includes **routers, switches, desktops, servers, and even security devices.**



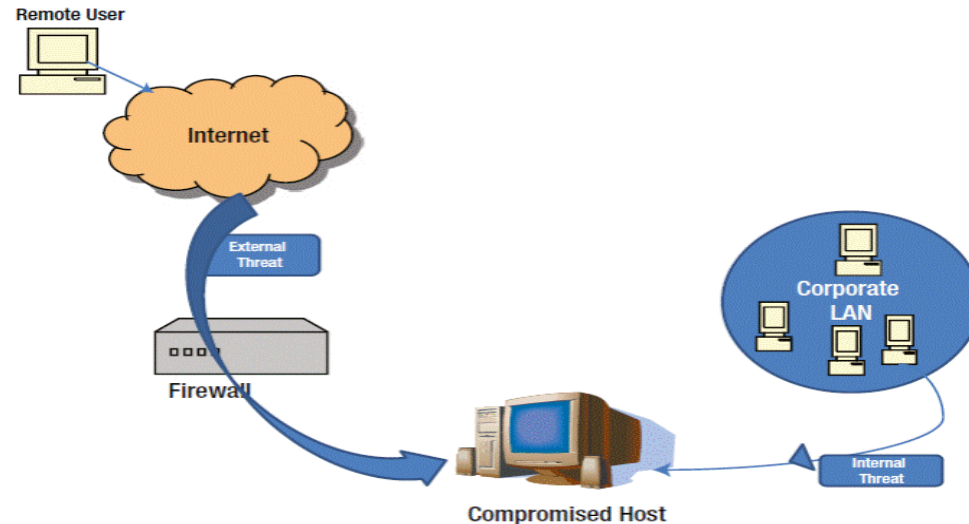
Threats

- A threat is an event that can take advantage of vulnerability and cause a negative impact on the network.
- Potential threats to the network need to be identified, and the related vulnerabilities need to be addressed to minimize the threat risk.



Types of Threats

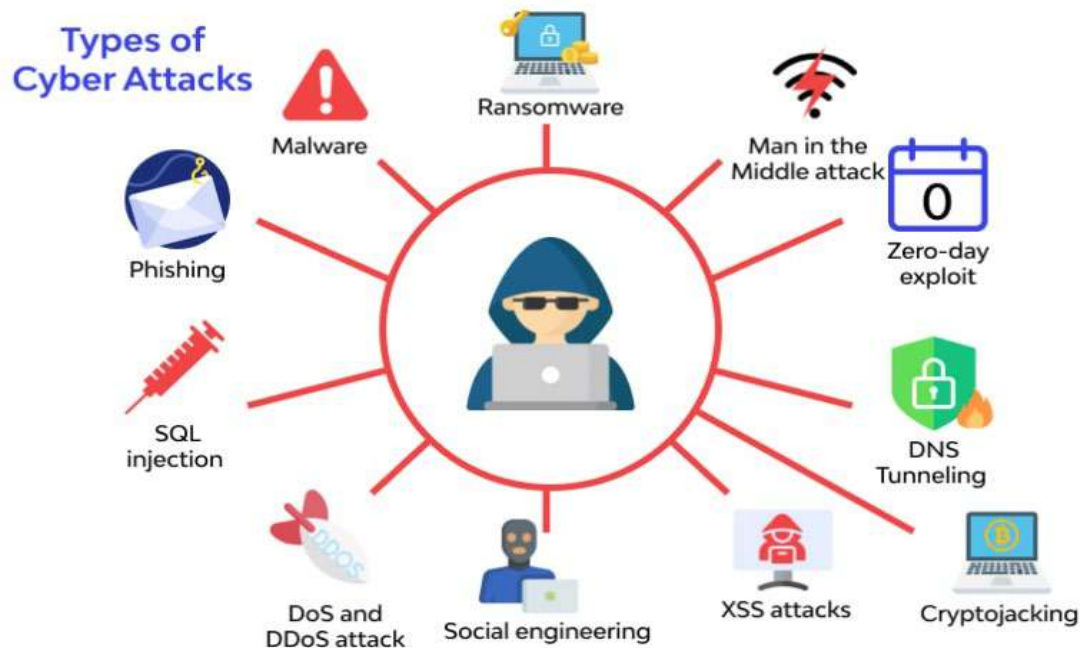
- **External threats** can arise from individuals or organizations working outside of a company.
- They do not have authorized access to the computer systems or network and work mainly from the Internet or dial-up access servers.



- **Internal threats** occur when someone has authorized access to the network with either an account on a server or physical access to the network.
- According to the FBI, internal access and misuse account for 60-80% of the reported incidents.

Attacks

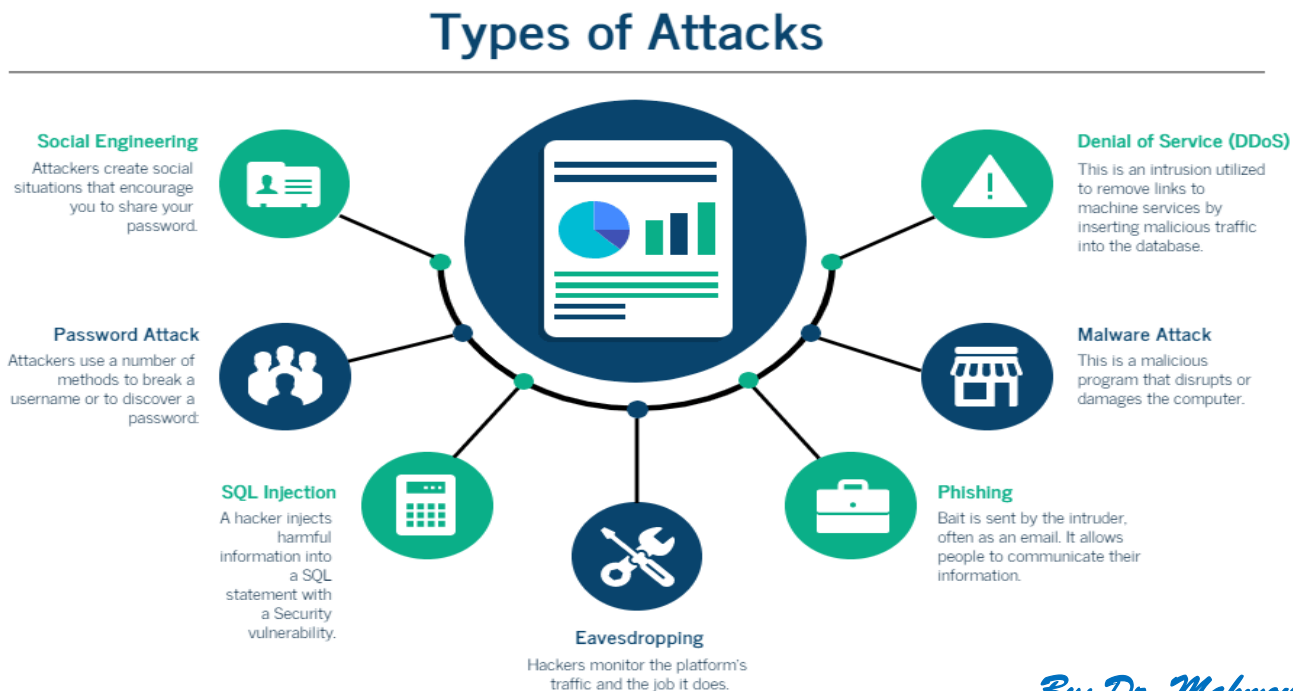
- **Attacks:** The threats use various **tools, scripts, and programs** to launch attacks against networks and network devices.
- Typically, the network devices under attack are the **endpoints**, such as servers and desktops.



Types of Attacks

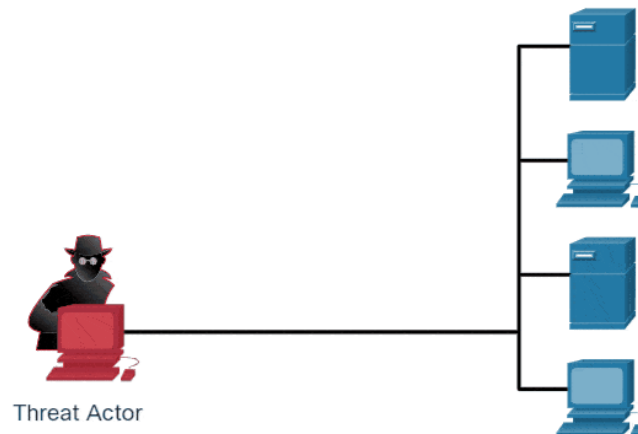
The primary classes of attacks exist are:

- Reconnaissance
- Access
- Denial of Service (DoS)
- Malwares
- Social Engineering



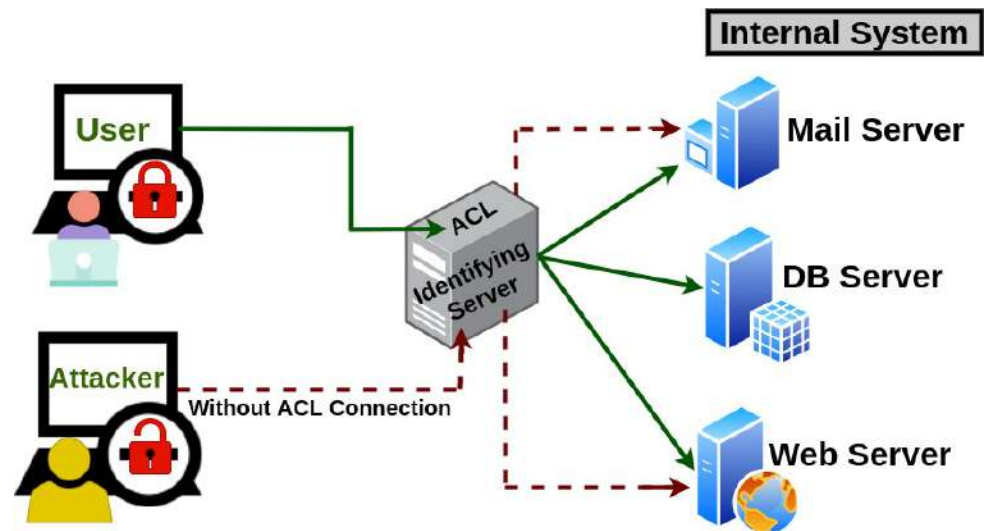
Reconnaissance Attack

- **Reconnaissance attack** is a kind of **information gathering** on network systems and services. This enables the attacker to discover vulnerabilities or weaknesses in the network
- In most cases, it precedes an actual access or **denial-of-service (DoS)** attack.
- Reconnaissance is somewhat analogous to a thief casing a neighborhood for vulnerable homes to break into, such as an unoccupied residence, easy-to-open doors, or open windows.



Access Attack

- Access attacks exploit known vulnerabilities in authentication services, FTP services, and web services to gain entry to web accounts, confidential databases, and other sensitive information.
- Access attacks can be classified into four types:
 - ☐ Password attacks
 - ☐ Trust exploitation
 - ☐ Port redirection
 - ☐ Man-in-the middle



Broken Access Control

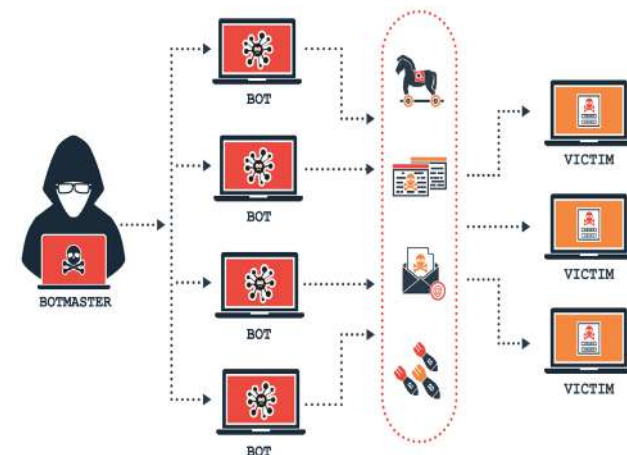
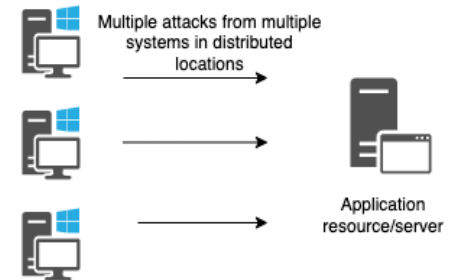
Denial of Service Attack

- **Denial of Service (DoS)** attacks are the most publicized form of attack and among the most difficult to eliminate.
- DoS attacks involve either **crashing** the system or **slowing it down** to the point that it is unusable.
- Numerous sites have been brought down by denial-of-service attacks, in which the attacker floods the site with traffic, rendering it unable to respond to legitimate queries.
- **DDoS Attack** means "**Distributed Denial-of-Service (DDoS) Attack**" is similar to a DoS attack but originates from multiple, coordinated sources. For example, an attacker builds a network of infected hosts, known as **zombies**. A network of zombies is called a **botnet**. The threat actor uses a command and control (CnC) program to instruct the botnet of zombies to carry out a DDoS attack

DoS attack



DDoS attack



Malware

- Malware is short for **malicious code**. It is code designed to damage, disrupt, steal, or inflict other “bad” or illegitimate actions on data, hosts, or networks.
- End devices are especially prone to malware attacks.
- The three most common types of malware are:
 - ☐ Virus
 - ☐ worm
 - ☐ Trojan horse

Types of malware



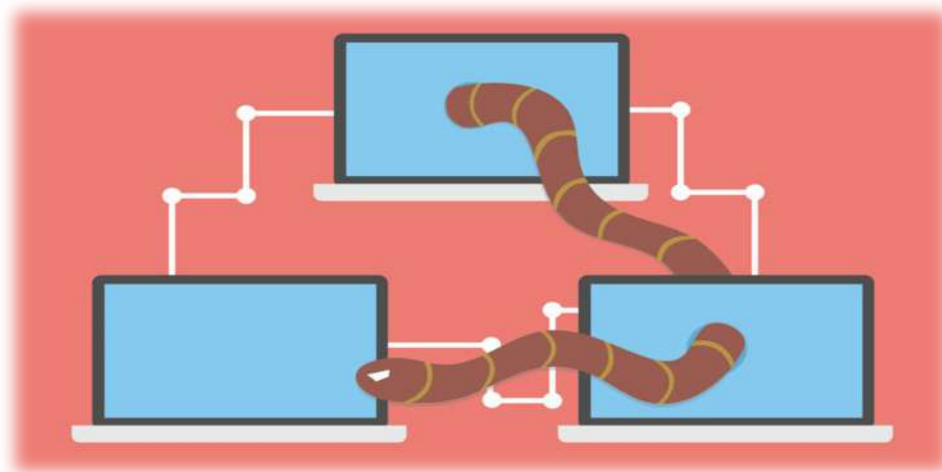
Viruses

- A virus is a type of malware that spreads by **inserting a copy of itself into another program**. After the program is run, viruses then spread from one computer to another, infecting the computers. Most viruses require human help to spread.
- Viruses can also be programmed to mutate to avoid detection.
- Most viruses are now spread by USB memory drives, CDs, DVDs, network shares, and email.



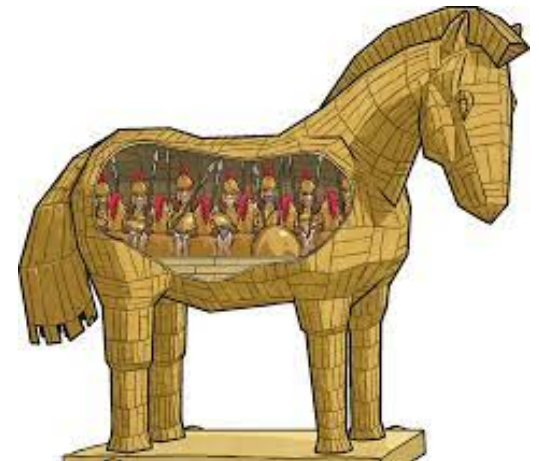
Worms

- Computer worms are like viruses because they replicate and can cause the same damage. Specifically, worms replicate themselves by **independently exploiting vulnerabilities in networks**.
- Worms can slow down networks as they spread from system to system.
- Whereas a **virus requires a host program** to run, **worms can run by themselves**. Other than the initial infection, worms no longer require user participation.



Trojan Horses

- Trojan horse malware is software that **appears legitimate**, but it contains malicious code that exploits the privileges of the user who runs it.
- Often, Trojans are found **attached to online games**. Users are commonly tricked into loading and executing the Trojan horse on their systems. While playing the game, the user will not notice a problem. In the background, the Trojan horse has been installed on the user's system. The malicious code from the Trojan horse continues operating even after the game has been closed.
- The Trojan horse concept is flexible. It can cause immediate damage, provide remote access to the system, or access through a back door. It can also perform actions as instructed remotely, such as "send me the password file once per week."



Ransomware

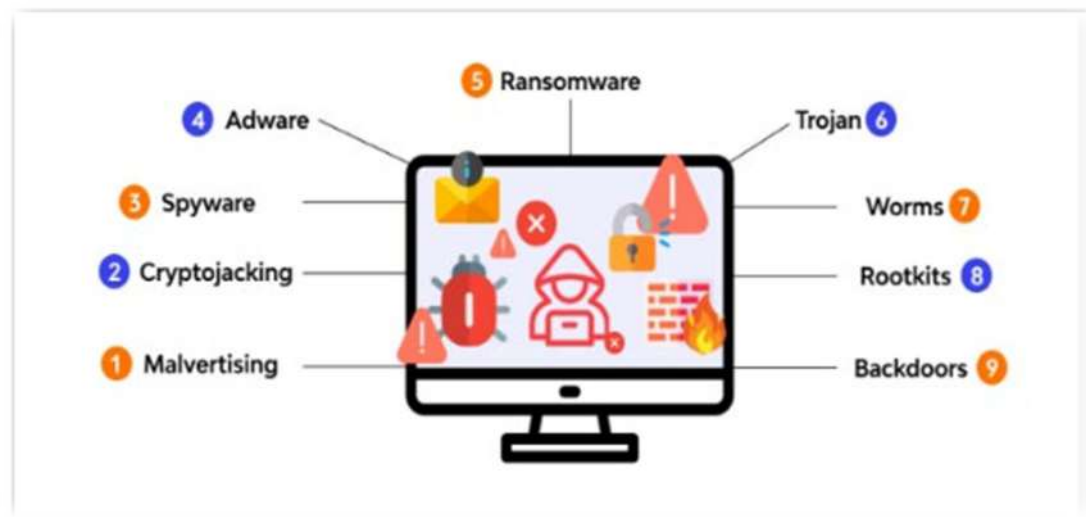
Currently, the most dominant malware is ransomware.

- Ransomware is malware that **denies access** to the infected computer system or its data. The cybercriminals then **demand payment** to release the computer system.
- Ransomware has evolved to become the most **profitable malware** type in history.
- Ransomware frequently **uses an encryption algorithm** to encrypt system files and data.
- Payments are typically made in **Bitcoin** because Bitcoin users can remain anonymous.



Other Types of Malware

Type of Malware	Description
Spyware	Used to gather information about a user and send the information to another entity without the user's consent. Spyware can be a system monitor, Trojan horse, Adware, tracking cookies, and key loggers.
Adware	Displays annoying pop-ups to generate revenue for its author. The malware may analyze user interests by tracking the websites visited. It can then send pop-up advertising pertinent to those sites.
Phishing	Attempts to convince people to disclose sensitive information. Examples include receiving an email from their bank asking users to divulge their account and PIN numbers.
Backdoors and Rootkits	Installed on a compromised system. After it is installed, it continues to hide its intrusion and provide privileged access to the threat actor.



Social Engineering

Social engineering is a completely **non-technical** means for a criminal to gather information on a target. Social engineering is an attack that attempts to manipulate individuals into performing actions or telling confidential information.

Social engineers often rely on people's willingness to be helpful but also target on people's weaknesses.

Social Engineering Tactics to Watch For

Knowing the red flags can help you avoid becoming a victim.



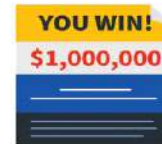
Your 'friend' sends you a strange message.



Your emotions are heightened.



The request is urgent.



The offer feels too good to be true.



You're receiving help you didn't ask for.



The sender can't prove their identity.

Hi this is Amy from the help desk. We need to upgrade the software on your computer after work hours. What is your user ID and password? You can change the password tomorrow when you log in.

Ok, my user ID and password are...

Social Engineer

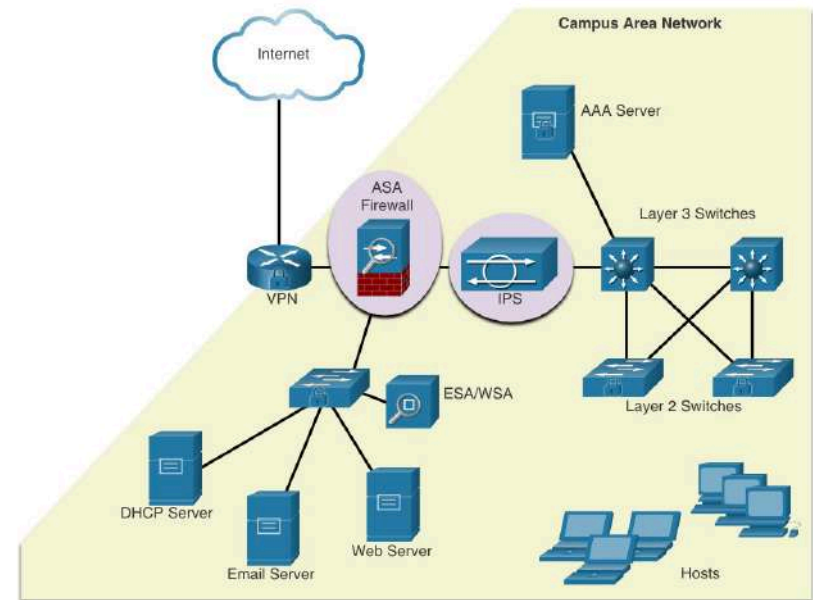
Unsuspecting Employee at XYZ Corporation

Network Attack Mitigations

To mitigate network attacks, you must first secure devices including routers, switches, servers, and hosts. Most organizations employ a **defense-in-depth approach** (also known as a layered approach) to security. This requires a combination of networking devices and services working together.

Several security devices and services are implemented to protect an organization's users and assets against TCP/IP threats:

- VPN
- Firewall
- IPS/IDS
- ESA/WSA
- AAA Server





Information Security

Information security protects information and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction. The CIA Triad serves as a conceptual foundation for the field.



The CIA Triad consists of three components of information security:

- **Confidentiality** - Only authorized individuals, entities, or processes can access sensitive information.
- **Integrity** - This refers to the protection of data from unauthorized alteration.
- **Availability** - Authorized users must have uninterrupted access to the network resources and data that they require

Creating More Experts

Cybersecurity Certifications

Industry Certifications

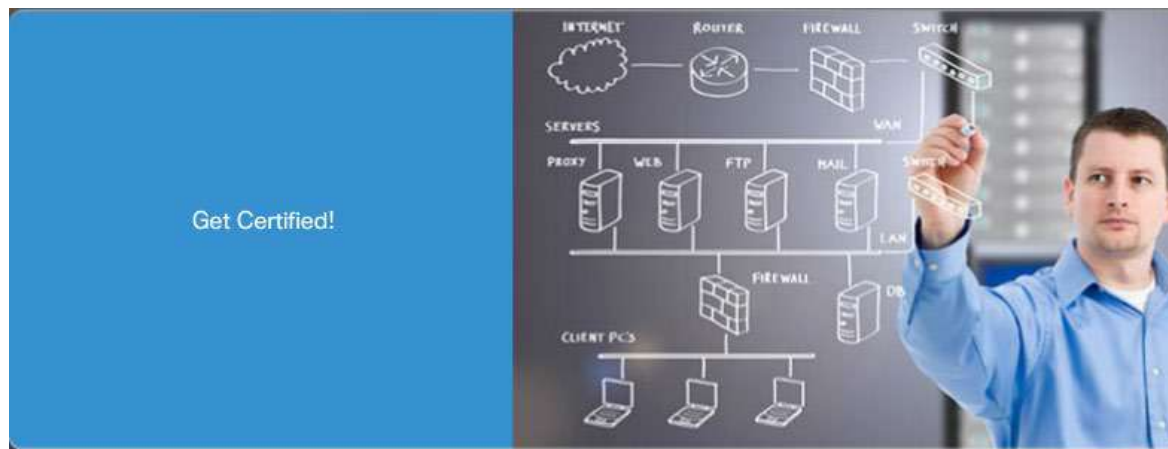
In a world of cybersecurity threats, there is a great need for skilled and knowledgeable information security professionals. The IT industry established standards for cybersecurity specialists to obtain professional certifications that provide proof of skills, and knowledge level.

- **CompTIA Security+** - Security+ is a CompTIA-sponsored testing program that certifies the competency of IT administrators in information assurance.
- **EC-Council Certified Ethical Hacker (CEH)** – CEH is an intermediate-level certification asserts that cybersecurity specialists holding this credential possess the skills and knowledge for various hacking practices.
- **SANS GIAC Security Essentials (GSEC)** - The GSEC certification is a good choice for an entry-level credential for cybersecurity specialists who can demonstrate that they understand security terminology and concepts and have the skills and expertise required for “hands-on” security roles. The SANS GIAC program offers a number of additional certifications in the fields of security administration, forensics, and auditing.

Creating More Experts

Cybersecurity Certifications (Cont.)

- **(ISC)² Certified Information Systems Security Professional (CISSP)** - The CISSP certification is a vendor-neutral certification for those cybersecurity specialists with a great deal of technical and managerial experience. It is also formally approved by the U.S. Department of Defense (DoD) and is a globally recognized industry certification in the security field.
- **ISACA Certified Information Security Manager (CISM)** – Cybersecurity specialists responsible for managing, developing and overseeing information security systems at the enterprise level or for those developing best security practices can qualify for CISM.



Creating More Experts

Cybersecurity Certifications (Cont.)

Company-Sponsored Certifications - Another important credential for cybersecurity specialists is company-sponsored certifications. These certifications measure knowledge and competency in installing, configuring, and maintaining vendor products. Cisco and Microsoft are examples of companies with certifications that test knowledge of their products. Click [here](#) to explore the matrix of the Cisco certifications shown in the figure.

Cisco Certified Network Associate Security (CCNA Security) - The CCNA Security certification validates that a cybersecurity specialist has the knowledge and skills required to secure Cisco networks.

Cisco Certifications				
	Entry	Associate	Professional	Expert
Architect				CCAr Architect
Cloud		CCNA Cloud	CCNP Cloud	
Collaboration		CCNA Collaboration	CCNP Collaboration	CCIE Collaboration
Data Center		CCNA Data Center	CCNP Data Center	CCIE Data Center
Design	CCENT	CCDA	CCDP	CCDE
Industrial / IoT		CCNA Industrial		
Routing & Switching	CCENT	CCNA Routing & Switching	CCNP Routing & Switching	CCIE Routing & Switching
Security	CCENT	CCNA Security	CCNP Security	CCIE Security
Service Provider		CCNA SP	CCNP SP	CCIE SP
Wireless	CCENT	CCNA Wireless	CCNP Wireless	CCIE Wireless
Other Certifications	Certified Technician			
Specialist	Business	Data Center	Internet of Things	Network Programmability
	Security	Operating System Software	Service Provider	Collaboration

By: Dr. Mahmoud Shuker Mahmoud

Cybersecurity Certifications (Cont.)

How to Become a Cybersecurity Specialist

Cybersecurity specialists must be able to respond to threats as soon as they occur. This means that the working hours can be somewhat unconventional. Cybersecurity specialists also analyze policy, trends, and intelligence to understand how cybercriminals think. Many times, this may involve a large amount of detective work. Here is good advice for becoming a cybersecurity specialist:



- **Study:** Learn the basics by completing courses in IT. Be a life-long learner. Cybersecurity is an ever-changing field, and cybersecurity specialists must keep up.
- **Pursue Certifications:** Industry and company-sponsored certifications from organizations such as Microsoft and Cisco prove one possesses the knowledge needed to seek employment as a cybersecurity specialist.
- **Pursue Internships:** Seeking a security internship as a student can lead to opportunities.
- **Join Professional Organizations:** Join computer security organizations, attend meetings and conferences, and join forums and blogs to gain knowledge from the experts.